

DOCKER ASSIGNMENT

1. **Docker installation:** Install Docker on your local system (Windows, macOS, or Linux).

Verify the installation by running the `docker --version` and `docker info` commands.

```
oot@LAPTOP-HPU0TJF1:~# docker --version
docker version 29.1.3, build 29.1.3-0ubuntu4.1
oot@LAPTOP-HPU0TJF1:~# docker info
Client:
Version:      29.1.3
Context:      default
Debug Mode:   false
Plugins:
  compose: Docker Compose (Docker Inc.)
    Version:  2.40.3+ds1-0ubuntu1
    Path:     /usr/libexec/docker/cli-plugins/docker-compose
  trust: Manage trust on Docker images (Docker Inc.)
    Version:  29.1.3
    Path:     /usr/libexec/docker/cli-plugins/docker-trust
Server:
Containers: 1
  Running: 1
  Paused: 0
  Stopped: 0
Images: 2
Server Version: 29.1.3
Storage Driver: overlayfs
  driver-type: io.containerd.snapshotter.v1
Logging Driver: json-file
Cgroup Driver: systemd
Cgroup Version: 2
Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
CDI spec directories:
  /etc/cdi
  /var/run/cdi
Swarm: inactive
Runtimes: io.containerd.runc.v2 runc
Default Runtime: runc
Init Binary: docker-init
containerd version:
```

2. **Working with Docker images:** Search for the official Ubuntu image on Docker Hub, pull it to your local system using `docker pull`, and list all available images on your system using `docker images`.



ubuntu
Docker Official Image · 1B+ · 10K+

Ubuntu is a Debian-based Linux operating system based on free software.

OPERATING SYSTEMS

[view](#)
[Tags](#)

Quick reference

- Maintained by:** [Canonical](#)
- Where to get help:** [the Docker Community Slack](#), [Server Fault](#), [Unix & Linux](#), or [Stack Overflow](#)

Supported tags and respective Dockerfile links

- 22.04 [jammy-20260627](#) [jammy](#)
- 24.04 [noble-20260610](#) [noble](#)
- 26.04 [resolute-20260707](#) [resolute](#) [latest](#) [rolling](#)
- 26.10 [stonking-20260705](#) [stonking](#) [devel](#)

```

root@LAPTOP-HPUOTJF1:~# docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
ed819469700f: Pull complete
a3679419df18: Pull complete
e16351a257e4: Download complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest
root@LAPTOP-HPUOTJF1:~# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
nginx:latest	b5a9a3cfc86b	241MB	66MB	U
ubuntu:latest	3131b4cc82a7	161MB	45.3MB	

3. **Creating a Docker container:** Run a Docker container using the Ubuntu image, and execute a command inside the container (e.g., echo "Hello, Docker!").

```

root@LAPTOP-HPUOTJF1:~# docker run -d -P --name ubuntu-cont ubuntu
6e1e3032fa2f6ca7e111132346ac7849e712b080d06977f5d07d1e2b0681ccb6
root@LAPTOP-HPUOTJF1:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
1ff04b87dfd5	nginx:latest	"/docker-entrypoint..."	4 hours ago	Up 4 hours	0.0.0.0:32768->80/tcp, [::]:32768->80/tcp	cloud_file_copy
root@1ff04b87dfd5:/#		docker exec -it 1ff04b87dfd5 /bin/bash				

```

root@1ff04b87dfd5:/# echo "Hello, Docker"
Hello, Docker
root@1ff04b87dfd5:/#

```

4. **Inspect the Ubuntu Docker container to ensure it is in the "created" state only.**

```

root@LAPTOP-HPUOTJF1:~# docker ps -a

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
2916347ea483	ubuntu	"/bin/bash"	19 seconds ago	Created		ubuntu-contai
6e1e3032fa2f	ubuntu	"/bin/bash"	5 minutes ago	Exited (0) 5 minutes ago		ubuntu-cont
1ff04b87dfd5	nginx:latest	"/docker-entrypoint..."	4 hours ago	Up 4 hours	0.0.0.0:32768->80/tcp, [::]:32768->80/tcp	cloud_file_co

```

root@LAPTOP-HPUOTJF1:~# docker inspect 2916347ea483
[
  {
    "Id": "2916347ea483fef5df8d2cc6b2726d0d64e59d7e052cd251c064cc7f102e9ffc",
    "Created": "2026-07-21T11:04:34.690861683Z",
    "Path": "/bin/bash",
    "Args": [],
    "State": {
      "Status": "created",
      "Running": false,
      "Paused": false,
      "Restarting": false,

```

5. **Execute the Ubuntu Docker container and confirm that the container's state is "running."**

```

root@LAPTOP-HPUOTJF1:~# docker inspect 1ff04b87dfd5
[
  {
    "Id": "1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88",
    "Created": "2026-07-21T07:15:14.759809887Z",
    "Path": "/docker-entrypoint.sh",
    "Args": [
      "nginx",
      "-g",
      "daemon off;"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,

```

- Launch an HTTP container, verify its state, and then remove the container and image from your local system.

```

root@LAPTOP-HPUOTJF1:~# docker run -d -P --name http-container httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
39d55e966aeb: Pull complete
4f4fb70ef54: Pull complete
69b5b831a361: Pull complete
ebd81352b960: Pull complete
010f7d7e175f: Pull complete
507729654175: Download complete
c72742279dc1: Download complete
Digest: sha256:305fd8326a27a137fedb8900f26375699ab233cc37793f35cf5d7c65671fb252
Status: Downloaded newer image for httpd:latest
e7fc705c2cc58c66e10fe62bdd9f1ff8097413933aabf78ecdaf56cfb6800c9
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
e7fc705c2cc5   httpd      "httpd-foreground"      8 seconds ago Up 7 seconds  0.0.0.0:32769->80/tcp, [::]:32769->80/tcp  http-container
1ff04b87dfd5   nginx:late "/docker-entrypoint..." 4 hours ago   Up 4 hours    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp  cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker rm -f e7fc
e7fc
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
1ff04b87dfd5   nginx:late "/docker-entrypoint..." 4 hours ago   Up 4 hours    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp  cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker images
INFO In Use
IMAGE ID          DISK USAGE  CONTENT SIZE  EXTRA
httpd:latest      305fd8326a27 177MB         47.6MB    U
nginx:latest      b5a9a3cfc86b 241MB         66MB      U
ubuntu:latest     3131b4cc82a7 161MB         45.3MB    U
root@LAPTOP-HPUOTJF1:~# docker rmi httpd:latest
Untagged: httpd:latest
Deleted: sha256:305fd8326a27a137fedb8900f26375699ab233cc37793f35cf5d7c65671fb252
root@LAPTOP-HPUOTJF1:~# docker images
INFO In Use
IMAGE ID          DISK USAGE  CONTENT SIZE  EXTRA
nginx:latest      b5a9a3cfc86b 241MB         66MB      U
ubuntu:latest     3131b4cc82a7 161MB         45.3MB    U
root@LAPTOP-HPUOTJF1:~#

```

- Execute a command to display a filtered list of containers, showing only their short container IDs.

```

root@LAPTOP-HPUOTJF1:~# docker ps -qa
2916347ea483
6e1e3032fa2f
1ff04b87dfd5
root@LAPTOP-HPUOTJF1:~#

```

- Clone the lancachenet/ubuntu-nginx Git repository locally and build a Docker image from the provided Dockerfile. Be sure to add relevant tags when building the image.

```

root@LAPTOP-HPUOTJF1:~# git clone https://github.com/lancachenet/ubuntu-nginx.git
Cloning into 'ubuntu-nginx'...
remote: Enumerating objects: 149, done.
remote: Counting objects: 100% (57/57), done.
remote: Compressing objects: 100% (36/36), done.
remote: Total 149 (delta 15), reused 41 (delta 11), pack-reused 92 (from 1)
Receiving objects: 100% (149/149), 32.15 KiB | 2.47 MiB/s, done.
Resolving deltas: 100% (30/30), done.
root@LAPTOP-HPUOTJF1:~# ls
1ff                                DevOps_Project  Merge_demo      clone-remote-repo  merge_confilct_demo  revert_demo  student-repo
Amend_repo                        GitIgnore-Repo  amend-repo      free-for-dev        project               stash-demo   ubuntu-nginx
Cloud_Ethix_Copyfiles            GitWorking      ci-cd-project   githubaction-project  revert-demo           stash1-demo
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# ls
Dockerfile  README.md  goss.yaml  goss_wait.yaml  overlay  run-tests.sh

```

```

root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker build -t ubuntu-nginx .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 137.7kB
Step 1/7 : FROM lancachenet/ubuntu:latest
latest: Pulling from lancachenet/ubuntu
e5044da18586: Pulling fs layer
199f361bf106: Pulling fs layer
0e5f8475f38b: Pulling fs layer
2b7f00c642c2: Pulling fs layer
e5044da18586: Download complete
2b7f00c642c2: Download complete
199f361bf106: Download complete
0e5f8475f38b: Download complete

---> eb1421f302fc
Step 7/7 : EXPOSE 80
---> Running in f009dc981031
---> Removed intermediate container f009dc981031
---> 5102563991c6
Successfully built 5102563991c6
Successfully tagged ubuntu-nginx:latest
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
lancachenet/ubuntu:latest           f86a1b63836e                     248MB        61.4MB
nginx:latest                         b5a9a3cfc86b                     241MB        66MB         U
ubuntu-nginx:latest                 5102563991c6                     315MB        80.4MB
ubuntu:latest                       3131b4cc82a7                     161MB        45.3MB         U
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx#

```

- After building the image, run a Docker container using the image and check the webpage using the curl command to ensure it's accessible.

```

root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker images
INFO In Use
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
achalpadol/ubuntu:latest           3131b4cc82a7                     161MB        45.3MB         U
achalpadol/ubuntu:v1               3131b4cc82a7                     161MB        45.3MB         U
lancachenet/ubuntu:latest          f86a1b63836e                     248MB        61.4MB
nginx:latest                       b5a9a3cfc86b                     241MB        66MB         U
ubuntu-nginx:latest                5102563991c6                     315MB        80.4MB
ubuntu:latest                      3131b4cc82a7                     161MB        45.3MB         U
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker run -d -P --name webpage-cont ubuntu-nginx:latest
a5f773380c62a2a14dfcde8023ec0bb485674c20e6ac481a6d48df2124fca50e
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                               NAMES
a5f773380c62   ubuntu-nginx:latest                 "/bin/bash -e /init/..." 7 seconds ago  Up 6 seconds  0.0.0.0:32770->80/tcp, [::]:32770->80/tcp  webpage-cont
1ff04b87df05   nginx:latest                         "/docker-entrypoint..." 4 hours ago    Up 4 hours    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp  cloud_file_copy
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# curl http://localhost:32770
Welcome to the lancachenet/ubuntu-nginx docker container
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx#

```

- Create a DockerHub account, set up a repository, and tag the locally created Docker image using your_repo_name/application_name:version_number.

achalpadol/ubuntu

Created less than a minute ago · ☆0 · ↓0

Add a description

Add a category

- General
- Tags
- Image Management
- Collaborators
- Webhooks
- Settings

Tags

DOCKER SCOUT INACTIVE

```
sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb -> sha256:7c2884fd32770fc6c173b78e0dc2278a2851d89f5447919edbc45475ac55dd6a
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker tag ubuntu:latest achalpadol/ubuntu:v1
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx# docker push achalpadol/ubuntu:v1
The push refers to repository [docker.io/achalpadol/ubuntu]
a3679419df18: Layer already exists
ed819469700f: Layer already exists
v1: digest: sha256:7c2884fd32770fc6c173b78e0dc2278a2851d89f5447919edbc45475ac55dd6a size: 1392

Info -> Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb -> sha256:7c2884fd32770fc6c173b78e0dc2278a2851d89f5447919edbc45475ac55dd6a
root@LAPTOP-HPUOTJF1:~/ubuntu-nginx#
```

achalpadol/ubuntu

Last pushed less than a minute ago · Repository size: 39.7 MB · ☆0 · ↓0

Add a description

Add a category

- General
- Tags
- Image Management
- Collaborators
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Tags

DOCKER SCOUT INACTIVE
[Activate](#)

This repository contains 0 tag(s).

Tag	OS	Type	Pulled	Pushed
v1		Image	less than 1 day	less than a minute
latest		Image	less than 1 day	2 minutes

[See all](#)

11. . Launch an Ubuntu container, demonstrate that the escape sequence is functional, and reattach the container after detaching it with the escape sequence. Then, check the container's IP address.

- Launch ubuntu container and back to the terminal by using escape sequence ctrl+p and ctrl+q and then reattach then container

```
root@LAPTOP-HPUOTJF1:~# docker run -it --name ubuntu-demo1 ubuntu /bin/bash
root@233271ccc3ef:/# root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS          NAMES
233271ccc3ef   ubuntu   "/bin/bash"             About a minute ago    Up About a minute                ubuntu-demo1
root@LAPTOP-HPUOTJF1:~# docker attach ubuntu-demo1
root@233271ccc3ef:/# hostname -I
172.17.0.2
root@233271ccc3ef:/#
```

12. Use a Docker command to inspect the hostname and /etc/hosts file of the httpd container.

```
exit
root@LAPTOP-HPUOTJF1:~# docker run -d -p 80:80 --name httpd-cont httpd
Unable to find image 'httpd:latest' locally
latest: Pulling from library/httpd
010f7d7e175f: Pull complete
4f4fb70ef54: Pull complete
69b5b831a361: Pull complete
ebd81352b960: Pull complete
39d55e966aeb: Pull complete
c72742279dc1: Download complete
507729664175: Download complete
Digest: sha256:305fd8326a27a137fedb8900f26375699ab233cc37793f35cf5d7c65671fb252
Status: Downloaded newer image for httpd:latest
f6d1a4859e0ab879ebaec9f3fe93fb9134e808227fcf3a587283f6369e3edc4e
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS          NAMES
f6d1a4859e0a   httpd     "httpd-foreground"      5 seconds ago    Up 5 seconds    0.0.0.0:80->80/tcp, [::]:80->80/tcp    httpd-cont
root@LAPTOP-HPUOTJF1:~# docker inspect -f '{{.Config.Hostname}}' httpd-cont
f6d1a4859e0a
root@LAPTOP-HPUOTJF1:~# cat $(docker inspect -f '{{.HostsPath}}' httpd-cont)
127.0.0.1      localhost
::1            localhost ip6-localhost ip6-loopback
fe00::        ip6-localnet
ff00::        ip6-mcastprefix
ff02::1       ip6-allnodes
ff02::2       ip6-allrouters
172.17.0.2     f6d1a4859e0a
```

13. Run a Docker command to inspect a container created with the lancachenet/ubuntu-nginx repository, and monitor the container's resource usage and stats with the TOP command.

```
172.17.0.3      d881f8031d75
root@LAPTOP-HPUOTJF1:~# docker inspect webpage-cont
[
  {
    "Id": "a5f773380c62a2a14dfcde8023ec0bb485674c20e6ac481a6d48df2124fca50e",
    "Created": "2026-07-21T11:33:06.191906718Z",
    "Path": "/bin/bash",
    "Args": [
      "-e",
      "/init/entrypoint",
      "/init/supervisord"
    ],
    "State": {
      "Status": "running",
      "Running": true,
      "Paused": false,
      "Restarting": false,
      "OOMKilled": false,
      "Dead": false,

```

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
a5f773380c62	webpage-cont	0.02%	33.19MiB / 7.581GiB	0.43%	1.9kB / 791B	29.4MB / 28.7kB	15

```

^Croot@LAPTOP-HPUOTJF1:~# docker top webpage-cont
UID          PID         PPID         C            STIME       TTY          TIME         CMD
root         8148        8124         0            11:33       ?            00:00:00    /usr/bin/python3
root         8226        8148         0            11:33       ?            00:00:00    /bin/bash /opt/n
glnx/startnginx.sh
root         8227        8226         0            11:33       ?            00:00:00    nginx: master pr
ocess /usr/sbin/nginx -g daemon off;
www-data     8228        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8229        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8230        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8231        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8232        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8233        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8234        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8235        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8236        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8237        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8238        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
www-data     8239        8227         0            11:33       ?            00:00:00    nginx: worker pr
ocess
root@LAPTOP-HPUOTJF1:~# |

```

14. Start a new Nginx container and review the logs for both Nginx containers.

```

root@LAPTOP-HPUOTJF1:~# docker run -d -P --name nginx-cont1 nginx:latest
41befcc7c7078a9c5fc083b68ad7f91a5179b73c1913d1560ca12d290e1817d6
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
41befcc7c707   nginx:latest   "/docker-entrypoint...." 4 seconds ago  Up 4 seconds  0.0.0.0:32772->80/tcp, [::]:32772->80/tcp  nginx-cont1
d881f6051b75   httpd         "httpd-foreground"        10 minutes ago Up 10 minutes  0.0.0.0:32771->80/tcp, [::]:32771->80/tcp  http-cont
1b6d1bccb2c9   ubuntu        "bash"                    24 minutes ago Up 24 minutes  -                                ubuntu-demo
a5f977380c62   ubuntu-nginx:latest "/bin/bash -e /init/..." 31 minutes ago Up 31 minutes  0.0.0.0:32770->80/tcp, [::]:32770->80/tcp  webpage-cont
1ff04b87df65   nginx:latest   "/docker-entrypoint...." 5 hours ago    Up 5 hours    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp  cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker logs cloud_file_copy
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/21 07:15:15 [notice] 1#1: using the "epoll" event method
2026/07/21 07:15:15 [notice] 1#1: nginx/1.31.2
2026/07/21 07:15:15 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/21 07:15:15 [notice] 1#1: OS: Linux 6.6.114.1-microsoft-standard-WSL2
2026/07/21 07:15:15 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/07/21 07:15:15 [notice] 1#1: start worker processes
2026/07/21 07:15:15 [notice] 1#1: start worker process 29
2026/07/21 07:15:15 [notice] 1#1: start worker process 30
2026/07/21 07:15:15 [notice] 1#1: start worker process 31
2026/07/21 07:15:15 [notice] 1#1: start worker process 32
2026/07/21 07:15:15 [notice] 1#1: start worker process 33
2026/07/21 07:15:15 [notice] 1#1: start worker process 34
2026/07/21 07:15:15 [notice] 1#1: start worker process 35
2026/07/21 07:15:15 [notice] 1#1: start worker process 36
2026/07/21 07:15:15 [notice] 1#1: start worker process 37
2026/07/21 07:15:15 [notice] 1#1: start worker process 38
2026/07/21 07:15:15 [notice] 1#1: start worker process 39
2026/07/21 07:15:15 [notice] 1#1: start worker process 40

```

```

oot@LAPTOP-HPUOTJF1:~# docker logs nginx-cont1
docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
0-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
0-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
docker-entrypoint.sh: Configuration complete; ready for start up
026/07/21 12:04:41 [notice] 1#1: using the "epoll" event method
026/07/21 12:04:41 [notice] 1#1: nginx/1.31.2
026/07/21 12:04:41 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
026/07/21 12:04:41 [notice] 1#1: OS: Linux 6.6.114.1-microsoft-standard-WSL2
026/07/21 12:04:41 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
026/07/21 12:04:41 [notice] 1#1: start worker processes
026/07/21 12:04:41 [notice] 1#1: start worker process 29
026/07/21 12:04:41 [notice] 1#1: start worker process 30
026/07/21 12:04:41 [notice] 1#1: start worker process 31
026/07/21 12:04:41 [notice] 1#1: start worker process 32
026/07/21 12:04:41 [notice] 1#1: start worker process 33
026/07/21 12:04:41 [notice] 1#1: start worker process 34
026/07/21 12:04:41 [notice] 1#1: start worker process 35
026/07/21 12:04:41 [notice] 1#1: start worker process 36
026/07/21 12:04:41 [notice] 1#1: start worker process 37
026/07/21 12:04:41 [notice] 1#1: start worker process 38
026/07/21 12:04:41 [notice] 1#1: start worker process 39
026/07/21 12:04:41 [notice] 1#1: start worker process 40
oot@LAPTOP-HPUOTJF1:~# |

```

15. Examine system events, filtering them by date and the last 30 minutes. Additionally, apply two filters simultaneously using name and event.

```

root@LAPTOP-HPUOTJF1:~# docker events --since 30m
2026-07-21T11:54:10.406240831Z image pull httpd:latest (name=httpd)
2026-07-21T11:54:10.531494248Z container create d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (image=httpd, name=http-cont)
2026-07-21T11:54:10.69729520Z network connect 69bcd6cbe2ee95b53de01b7e5351eb2b23995972d7a071fb773351129c780412 (container=d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae, name=bridge, type=bridge)
2026-07-21T11:54:10.707743433Z container start d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (image=httpd, name=http-cont)
2026-07-21T11:55:46.403396577Z container exec_create: hostname d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (execID=4777c4c17fbae0c41814e0d50df771fc86b37ba4dd9e9a6710542f5ae5d2094d, image=httpd, name=http-cont)
2026-07-21T11:55:46.403960046Z container exec_start: hostname d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (execID=4777c4c17fbae0c41814e0d50df771fc86b37ba4dd9e9a6710542f5ae5d2094d, image=httpd, name=http-cont)
2026-07-21T11:55:46.443713592Z container exec_die d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (execID=4777c4c17fbae0c41814e0d50df771fc86b37ba4dd9e9a6710542f5ae5d2094d, exitCode=0, image=httpd, name=http-cont)
2026-07-21T11:56:01.441429119Z container exec_create: cat /etc/hosts d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (execID=2497a51589568dc2fc36f0073ea57b3905e016ef63824a62e42880f173d4f6dc, image=httpd, name=http-cont)
2026-07-21T11:56:01.441798394Z container exec_start: cat /etc/hosts d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (execID=2497a51589568dc2fc36f0073ea57b3905e016ef63824a62e42880f173d4f6dc, image=httpd, name=http-cont)
2026-07-21T11:56:01.473823815Z container exec_die d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (execID=2497a51589568dc2fc36f0073ea57b3905e016ef63824a62e42880f173d4f6dc, exitCode=0, image=httpd, name=http-cont)
2026-07-21T12:01:20.323226933Z container top a5f773380c62a2a14dfdc8023ec0bb485674c20e6ac481a6d48df2124fca50e (image=ubuntu-nginx:latest, maintainer=LanCache.Net Team <team@lancache.net>, name=webpage-cont, org.opencontainers.image.version=24.04)
2026-07-21T12:04:41.386710709Z container create 41befcc7c7078a9c5fc083b68ad7f91a5179b73c1913d1560ca12d290e1817d6 (image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=nginx-cont1)
2026-07-21T12:04:41.542243227Z network connect 69bcd6cbe2ee95b53de01b7e5351eb2b23995972d7a071fb773351129c780412 (container=41befcc7c7078a9c5fc083b68ad7f91a5179b73c1913d1560ca12d290e1817d6, name=bridge, type=bridge)
2026-07-21T12:04:41.55329927Z container start 41befcc7c7078a9c5fc083b68ad7f91a5179b73c1913d1560ca12d290e1817d6 (image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=nginx-cont1)

root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# docker events --since "2026-07-21T00:00:00"
2026-07-21T07:15:14.798236122Z container create 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:15:14.798495810Z network connect 69bcd6cbe2ee95b53de01b7e5351eb2b23995972d7a071fb773351129c780412 (container=1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88, name=bridge, type=bridge)
2026-07-21T07:15:14.991353865Z container start 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:16:48.4442258188Z container exec_create: /bin/bash 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=cde3029b4c9a43ae5ba5a758fe23dc86309b575114288b04399757c3fc33c9e1, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:16:48.4443807868Z container exec_start: /bin/bash 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=cde3029b4c9a43ae5ba5a758fe23dc86309b575114288b04399757c3fc33c9e1, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:16:56.122985154Z container exec_die 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=cde3029b4c9a43ae5ba5a758fe23dc86309b575114288b04399757c3fc33c9e1, exitCode=9, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:18:10.1090803014Z container exec_create: /bin/bash 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=2035942d01dd3ac2989ace75937b71a6320c1105ff374bb445d174f26cd7b540, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:18:10.110769765Z container exec_start: /bin/bash 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=2035942d01dd3ac2989ace75937b71a6320c1105ff374bb445d174f26cd7b540, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:19:34.384579298Z container exec_die 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=2035942d01dd3ac2989ace75937b71a6320c1105ff374bb445d174f26cd7b540, exitCode=0, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:23:02.631601892Z container extract-to-dir 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)
2026-07-21T07:23:17.301740933Z container exec_create: /bin/bash 1ff04b87dfd56c6bf2e193810432698c0b355b2395ed482addcbcac32d67eb88 (execID=9e0c7e33508f37c5b5408b2a309b575114288b04399757c3fc33c9e1, image=nginx:latest, maintainer=NGINX Docker Maintainers <docker-maint@nginx.com>, name=cloud_file_copy)

root@LAPTOP-HPUOTJF1:~# docker events --since 30m \
--filter container=http-cont \
--filter event=start
2026-07-21T11:54:10.707743433Z container start d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (image=httpd, name=http-cont)

```

16. Create a Docker container using the Ubuntu image, assign it a meaningful name, and set the hostname to "cloudeithix.com." Stop or kill the container, check its exit code, and examine the container's log for more information.

```

ubuntu-cloud
root@LAPTOP-HPUOTJF1:~# docker exec -it ubuntu-cloud /bin/bash
root@cloudeithix:/# hostname
cloudeithix.com

2026-07-21T11:54:10.707743433Z container start d881f6051b755b31b4bf14503f04ad1ee0ecbaddc77c5e0a912dadcd7d44e61ae (image=httpd, name=http-cont)
root@LAPTOP-HPUOTJF1:~# docker run -dit \
--name ubuntu-cloud \
--hostname cloudeithix.com \
ubuntu bash
dcdbfed91d20 ubuntu 4 seconds ago Up 4 seconds 0.0.0.0:32772->80/tcp, [::]:32772->80/tcp NAMES ubuntu-cloud
41befcc7c707 nginx:latest 16 minutes ago Up 16 minutes 0.0.0.0:32771->80/tcp, [::]:32771->80/tcp nginx-cont1
d881f6051b75 httpd 27 minutes ago Up 27 minutes 0.0.0.0:32771->80/tcp, [::]:32771->80/tcp http-cont
1b6d1bccb2c0 ubuntu 41 minutes ago Up 41 minutes 0.0.0.0:32770->80/tcp, [::]:32770->80/tcp ubuntu-demo
a5f773380c62 ubuntu-nginx:latest 48 minutes ago Up 48 minutes 0.0.0.0:32770->80/tcp, [::]:32770->80/tcp webpage-cont
1ff04b87dfd5 nginx:latest 5 hours ago Up 5 hours 0.0.0.0:32768->80/tcp, [::]:32768->80/tcp cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker stop dcd
dcd
root@LAPTOP-HPUOTJF1:~# docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS
dcdbfed91d20 ubuntu "bash" About a minute ago Exited (137) 5 seconds ago
ubuntu-cloud

Hello from CloudEthix
root@LAPTOP-HPUOTJF1:~# docker logs ubuntu-log
Hello from CloudEthix
root@LAPTOP-HPUOTJF1:~#

```

17. Instantiate and start an Ubuntu container with a random name, then rename it to "Ubuntu01." Launch another container named "Ubuntu02" and check the hostname of

both containers. Pause and un-pause the "Ubuntu02" container, and stop, start, and restart the "Ubuntu01" container. Inspect the stats and system events of both containers, then kill and delete them, ensuring that directories created by the containers are removed.

```
root@LAPTOP-HPUOTJF1:~# docker run -dit ubuntu bash
ac6a1c7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209
root@LAPTOP-HPUOTJF1:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ac6a1c7a8a13	ubuntu	"bash"	4 seconds ago	Up 3 seconds		pedantic_he

```
root@LAPTOP-HPUOTJF1:~# docker rename pedantic_hertz Ubuntu01
root@LAPTOP-HPUOTJF1:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
ac6a1c7a8a13	ubuntu	"bash"	About a minute ago	Up About a minute		Ubuntu01

```
root@LAPTOP-HPUOTJF1:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b12d575cf077	ubuntu	"bash"	4 seconds ago	Up 4 seconds		Ubuntu02
ac6a1c7a8a13	ubuntu	"bash"	3 minutes ago	Up 3 minutes		Ubuntu01

```
copy
root@LAPTOP-HPUOTJF1:~# docker exec Ubuntu01 hostname
ac6a1c7a8a13
root@LAPTOP-HPUOTJF1:~# docker exec Ubuntu02 hostname
b12d575cf077
root@LAPTOP-HPUOTJF1:~# |
```

```
Ubuntu02
root@LAPTOP-HPUOTJF1:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b12d575cf077	ubuntu	"bash"	2 minutes ago	Up 2 minutes (Paused)		Ubuntu
ac6a1c7a8a13	ubuntu	"bash"	5 minutes ago	Up 5 minutes		Ubuntu

```
root@LAPTOP-HPUOTJF1:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b12d575cf077	ubuntu	"bash"	2 minutes ago	Up 2 minutes		Ubuntu02
ac6a1c7a8a13	ubuntu	"bash"	6 minutes ago	Up 6 minutes		Ubuntu01

```
root@LAPTOP-HPUOTJF1:~# docker stop ac6
root@LAPTOP-HPUOTJF1:~# ^C
root@LAPTOP-HPUOTJF1:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b12d575cf077	ubuntu	"bash"	4 minutes ago	Up 4 minutes		Ubuntu02
dcdbfed91d20	ubuntu	"bash"	21 minutes ago	Up 16 minutes		ubuntu-clou
41befcc7c707	nginx:latest	"/docker-entrypoint..."	38 minutes ago	Up 38 minutes	0.0.0.0:32772->80/tcp, [::]:32772->80/tcp	nginx-cont1
4881f6051b75	httpd	"httpd-foreground"	49 minutes ago	Up 49 minutes	0.0.0.0:32771->80/tcp, [::]:32771->80/tcp	http-cont
1b6d1bccb2c0	ubuntu	"bash"	About an hour ago	Up About an hour		ubuntu-demo
a5f773380c62	ubuntu-nginx:latest	"/bin/bash -e /init/..."	About an hour ago	Up About an hour	0.0.0.0:32770->80/tcp, [::]:32770->80/tcp	webpage-con
1ff04b87dfd5	nginx:latest	"/docker-entrypoint..."	5 hours ago	Up 5 hours	0.0.0.0:32768->80/tcp, [::]:32768->80/tcp	cloud_file_

```
copy
root@LAPTOP-HPUOTJF1:~# docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b12d575cf077	ubuntu	"bash"	4 minutes ago	Up 4 minutes		Ubuntu02
ac6a1c7a8a13	ubuntu	"bash"	8 minutes ago	Exited (137) 18 seconds ago		Ubuntu01

```
cloud_file_copy
root@LAPTOP-HPUOTJF1:~# docker start ac6
ac6
root@LAPTOP-HPUOTJF1:~# docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
b12d575cf077	ubuntu	"bash"	6 minutes ago	Up 6 minutes		Ubuntu02
ac6a1c7a8a13	ubuntu	"bash"	9 minutes ago	Up 7 seconds		Ubuntu01
dcdbfed91d20	ubuntu	"bash"	23 minutes ago	Up 17 minutes		ubuntu-clou

```
root@LAPTOP-HPUOTJF1:~# docker restart Ubuntu01
Ubuntu01
root@LAPTOP-HPUOTJF1:~# |
```

Stats of the both containers

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
ac6a1c7a8a13	Ubuntu01	0.00%	884KiB / 7.581GiB	0.01%	726B / 126B	0B / 0B	1
b12d575cf077	Ubuntu02	0.00%	952KiB / 7.581GiB	0.01%	1.19kB / 126B	0B / 0B	1

Events of the ubuntu01

Run this cmd on the another terminal then check the events on the first terminal.

```

acna1@LAPTOP-HPUOTJF1:~$ sudo -l
[sudo: authenticate] Password:
root@LAPTOP-HPUOTJF1:~# docker stop Ubuntu01
^Croot@LAPTOP-HPUOTJF1:~# docker start Ubuntu01
Ubuntu01
root@LAPTOP-HPUOTJF1:~# docker restart Ubuntu01
Ubuntu01
root@LAPTOP-HPUOTJF1:~# docker pause Ubuntu02
Ubuntu02
root@LAPTOP-HPUOTJF1:~# docker unpause Ubuntu02
Ubuntu02
root@LAPTOP-HPUOTJF1:~# docker restart Ubuntu02
Ubuntu02
root@LAPTOP-HPUOTJF1:~# █

```

```

root@LAPTOP-HPUOTJF1:~# docker events --filter container=Ubuntu01
2026-07-21T12:51:32.821834695Z container kill ac6alc7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209 (image=ubuntu, name=Ubuntu01, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
It is the world's most popular operating system across public clouds and OpenStack clouds.
It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04, signal=15)
2026-07-21T12:51:42.837129595Z container kill ac6alc7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209 (image=ubuntu, name=Ubuntu01, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

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It is the world's most popular operating system across public clouds and OpenStack clouds.
It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04, signal=9)
2026-07-21T12:51:43.009102324Z container stop ac6alc7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209 (image=ubuntu, name=Ubuntu01, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

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, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04)
2026-07-21T12:51:43.011616289Z container die ac6alc7a8a135b51ae2c79d5db99e247e1dda24d8ae0930937ab5c2c7ce03209 (execDuration=337, exitCode=137, image=ubuntu, name=Ubuntu01, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical)

```

Event of the ubuntu02

```

^Croot@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# docker events --filter container=Ubuntu02

2026-07-21T12:58:19.113279016Z container pause b12d575cf07732453db744952f2b728e358451e9074bc259adbc74206ba616b3 (image=ubuntu, name=Ubuntu02, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
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It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04)
2026-07-21T12:58:29.845691835Z container unpause b12d575cf07732453db744952f2b728e358451e9074bc259adbc74206ba616b3 (image=ubuntu, name=Ubuntu02, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

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It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04)
2026-07-21T12:58:36.867574193Z container kill b12d575cf07732453db744952f2b728e358451e9074bc259adbc74206ba616b3 (image=ubuntu, name=Ubuntu02, org.opencontainers.image.created=2026-07-13T16:06:30.499069+00:00, org.opencontainers.image.description=The Ubuntu container image maintained by Canonical

Ubuntu is a Debian-based Linux operating system that runs from the desktop to the cloud, to all your internet connected things.
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It is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.
Fast, secure and simple, Ubuntu powers millions of PCs worldwide.
, org.opencontainers.image.title=ubuntu, org.opencontainers.image.version=26.04)

```

- Run four HTTPD Docker containers with distinct, meaningful names, and apply restart policies (NO, On-Failure, Always, and Unless-Stopped) to each of the four containers, respectively. Demonstrate that the restart policies function as expected.

- Four containers are created with different name

```

4f2ed216677d01fb8e0ad38e62a990d5f91542b72b652c487181ef3a0233a0bc
root@LAPTOP-HPUOTJF1:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
4f2ed216677d	httpd	"httpd-foreground"	5 seconds ago	Up 4 seconds	0.0.0.0:32771->80/tcp, [::]:32771->80/tcp	httpd-unless
977d5e5c948f	httpd	"httpd-foreground"	23 seconds ago	Up 22 seconds	0.0.0.0:32770->80/tcp, [::]:32770->80/tcp	httpd-always
db96e974a76b	httpd	"httpd-foreground"	54 seconds ago	Up 53 seconds	0.0.0.0:32769->80/tcp, [::]:32769->80/tcp	httpd-onfailure
f67caab69cc5	httpd	"httpd-foreground"	About a minute ago	Up About a minute	0.0.0.0:32768->80/tcp, [::]:32768->80/tcp	httpd-no

```

root@LAPTOP-HPUOTJF1:~#

```

1. --restart=no

No restart policy will not start container automatically

```
root@LAPTOP-HPUOTJF1:~# docker stop httpd-no
httpd-no
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
4f2ed216677d   httpd     "httpd-foreground"       4 minutes ago Up About a minute    0.0.0.0:32771->80/tcp, [::]:32771->80/tcp    httpd-unless
977d5e5c948f   httpd     "httpd-foreground"       About a minute ago Up About a minute    0.0.0.0:32770->80/tcp, [::]:32770->80/tcp    httpd-always
db96e974a76b   httpd     "httpd-foreground"       About a minute ago Up About a minute    0.0.0.0:32769->80/tcp, [::]:32769->80/tcp    httpd-onfailure
```

2. --restart=on-failure

This policy only restarts the container if the main process exits with a **non-zero exit code** (an error).

```
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# docker stop httpd-onfailure
httpd-onfailure
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
4f2ed216677d   httpd     "httpd-foreground"       4 minutes ago Up 4 minutes    0.0.0.0:32771->80/tcp, [::]:32771->80/tcp    httpd-unless
977d5e5c948f   httpd     "httpd-foreground"       5 minutes ago Up 5 minutes    0.0.0.0:32770->80/tcp, [::]:32770->80/tcp    httpd-always
root@LAPTOP-HPUOTJF1:~# docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
4f2ed216677d   httpd     "httpd-foreground"       5 minutes ago Up 5 minutes    0.0.0.0:32771->80/tcp, [::]:32771->80/tcp    httpd-unless
977d5e5c948f   httpd     "httpd-foreground"       5 minutes ago Up 5 minutes    0.0.0.0:32770->80/tcp, [::]:32770->80/tcp    httpd-always
db96e974a76b   httpd     "httpd-foreground"       5 minutes ago Exited (0) 17 seconds ago                httpd-onfailure
```

will not restart it, because docker stop sends a graceful stop signal, resulting in exit code 0.

3. --restart=always

```
root@LAPTOP-HPUOTJF1:~# docker stop httpd-always
httpd-always
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
4f2ed216677d   httpd     "httpd-foreground"       10 minutes ago Up 10 minutes    0.0.0.0:32771->80/tcp, [::]:32771->80/tcp    httpd-unless
root@LAPTOP-HPUOTJF1:~# docker inspect -f '{{.HostConfig.RestartPolicy.Name}}'
docker: 'docker inspect' requires at least 1 argument

Usage:  docker inspect [OPTIONS] NAME|ID [NAME|ID...]

See 'docker inspect --help' for more information
root@LAPTOP-HPUOTJF1:~# docker inspect -f '{{.HostConfig.RestartPolicy.Name}}' httpd-always
always
root@LAPTOP-HPUOTJF1:~# sudo systemctl restart docker
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
4f2ed216677d   httpd     "httpd-foreground"       13 minutes ago Up 3 seconds    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp    httpd-unless
977d5e5c948f   httpd     "httpd-foreground"       13 minutes ago Up 3 seconds    0.0.0.0:32769->80/tcp, [::]:32769->80/tcp    httpd-always
root@LAPTOP-HPUOTJF1:~#
```

4. --restart=unless-stopped

Restart the container automatically unless it was manually stopped by the user

```
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
977d5e5c948f   httpd     "httpd-foreground"       17 minutes ago Up 4 seconds    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp    httpd-always
root@LAPTOP-HPUOTJF1:~# docker inspect -f '{{.HostConfig.RestartPolicy.Name}}' httpd-unless
unless-stopped
root@LAPTOP-HPUOTJF1:~#
```

19. Launch an NGINX container with a meaningful name and expose it on the host's port 80.

Create an "index.html" file containing the text "Hello there, Let's be the Team CloudEthiX," and copy the file to the container's "/usr/share/nginx/html/" location.

Access the container in a browser to verify that the webpage displays correctly.

```
unless-stopped
root@LAPTOP-HPUOTJF1:~# docker run -d --name nginx-web -p 80:80 nginx
1b33c35e955b7a877bea3ec1907332b117feaa7021c5717d105f1ec76735c2a8
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
1b33c35e955b   nginx     "/docker-entrypoint..." 4 seconds ago Up 3 seconds    0.0.0.0:80->80/tcp, [::]:80->80/tcp    nginx-web
977d5e5c948f   httpd     "httpd-foreground"       21 minutes ago Up 4 minutes    0.0.0.0:32768->80/tcp, [::]:32768->80/tcp    httpd-always
root@LAPTOP-HPUOTJF1:~# echo "Hello there, Let's be the Team CloudEthiX" > index.html
root@LAPTOP-HPUOTJF1:~# cat index.html
Hello there, Let's be the Team CloudEthiX
root@LAPTOP-HPUOTJF1:~# docker cp index.html nginx-web:/usr/share/nginx/html/index.html
Successfully copied 2.05kB to nginx-web:/usr/share/nginx/html/index.html
root@LAPTOP-HPUOTJF1:~# docker exec nginx-web cat /usr/share/nginx/html/index.html
Hello there, Let's be the Team CloudEthiX
root@LAPTOP-HPUOTJF1:~#
```



Hello there, Let's be the Team CloudEthiX

20. Pull the Redis image and tag it as "yourname_redis:version 1." Create two additional tags, "yourname_redis:version 2" and "yourname_redis:version 3." Run two Docker containers from the tagged images named "redis01" and "redis02," then delete the "yourname_redis:version 2" image.

```
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
achalpadol/ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
achalpadol/ubuntu:v1 3131b4cc82a7 161MB 45.3MB U
httpd:latest 305fd8326a27 177MB 47.6MB U
lancachenet/ubuntu:latest f86a1b63836e 248MB 61.4MB
nginx:latest b5a9a3cfc86b 241MB 66MB U
redis:latest 234c902a2db4 208MB 56.3MB
ubuntu-nginx:latest 5102563991c6 315MB 80.4MB U
ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
root@LAPTOP-HPUOTJF1:~# docker tag redis:latest achalpadol_redis:version1
root@LAPTOP-HPUOTJF1:~# docker images

IMAGE ID DISK USAGE CONTENT SIZE EXTRA
achalpadol/ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
achalpadol/ubuntu:v1 3131b4cc82a7 161MB 45.3MB U
achalpadol_redis:version1 234c902a2db4 208MB 56.3MB
httpd:latest 305fd8326a27 177MB 47.6MB U
lancachenet/ubuntu:latest f86a1b63836e 248MB 61.4MB
nginx:latest b5a9a3cfc86b 241MB 66MB U
redis:latest 234c902a2db4 208MB 56.3MB
ubuntu-nginx:latest 5102563991c6 315MB 80.4MB U
ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
root@LAPTOP-HPUOTJF1:~# docker tag redis:latest achalpadol_redis:version2
root@LAPTOP-HPUOTJF1:~# docker tag redis:latest achalpadol_redis:version3
root@LAPTOP-HPUOTJF1:~# docker image s
docker: unknown command: docker image s

Usage: docker image

Run 'docker image --help' for more information
root@LAPTOP-HPUOTJF1:~# docker images

IMAGE ID DISK USAGE CONTENT SIZE EXTRA
achalpadol/ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
achalpadol/ubuntu:v1 3131b4cc82a7 161MB 45.3MB U
achalpadol_redis:version1 234c902a2db4 208MB 56.3MB
achalpadol_redis:version2 234c902a2db4 208MB 56.3MB
achalpadol_redis:version3 234c902a2db4 208MB 56.3MB
httpd:latest 305fd8326a27 177MB 47.6MB U
ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# docker run -d --name redis01 achalpadol_redis:version1
b37b5a6ff268230f7a0f210832c2f78594b1e57e6763f014914d58cf6b5ef435
root@LAPTOP-HPUOTJF1:~# docker run -d --name redis02 achalpadol_redis:version3
ab6aa3634f8dd3edd34088cc68201fe165010515b69f6baadb9a9a2cfcf1468
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
ab6aa3634f80 achalpadol_redis:version3 "docker-entrypoint.s..." 15 seconds ago Up 14 seconds 6379/tcp redis02
b37b5a6ff268 achalpadol_redis:version1 "docker-entrypoint.s..." 45 seconds ago Up 44 seconds 6379/tcp redis01
1b33c35e985b nginx "/docker-entrypoint..." 6 minutes ago Up 6 minutes 0.0.0.0:80->80/tcp, [::]:80->80/tcp nginx-web
977d5e5c948f httpd "httpd-foreground" 28 minutes ago Up 11 minutes 0.0.0.0:32768->80/tcp, [::]:32768->80/tcp httpd-always
root@LAPTOP-HPUOTJF1:~# docker rmi achalpadol_redis:version2
Untagged: achalpadol_redis:version2
root@LAPTOP-HPUOTJF1:~# docker images

IMAGE ID DISK USAGE CONTENT SIZE EXTRA
achalpadol/ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
achalpadol/ubuntu:v1 3131b4cc82a7 161MB 45.3MB U
achalpadol_redis:version1 234c902a2db4 208MB 56.3MB U
achalpadol_redis:version3 234c902a2db4 208MB 56.3MB U
httpd:latest 305fd8326a27 177MB 47.6MB U
```

21. Clone the [GitHub - docker-library/httpd: Docker Official Image packaging for Apache HTTP Server](https://github.com/docker-library/httpd) Git repository locally and build the httpd Docker image from the Dockerfile at the "httpd/2.4/Dockerfile" path. Run and expose a Docker container from this image with a meaningful name on port 8181, then access the webpage in a browser. Afterward, tag the image as V1.

```
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# git clone https://github.com/docker-library/httpd.git
Cloning into 'httpd'...
remote: Enumerating objects: 1119, done.
remote: Counting objects: 100% (205/205), done.
remote: Compressing objects: 100% (94/94), done.
remote: Total 1119 (delta 144), reused 117 (delta 111), pack-reused 914 (from 4)
Receiving objects: 100% (1119/1119), 227.61 KiB | 547.00 KiB/s, done.
Resolving deltas: 100% (487/487), done.
root@LAPTOP-HPUOTJF1:~# ls
lff                                DevOps_Project  Merge_demo      clone-remote-repo  httpd            project          stash-demo      ubuntu-nginx
Amend_repo                        GitIgnore-Repo  amend-repo      free-for-dev       index.html       revert-demo      stash1-demo
Cloud_Ethix_Copyfiles            GitWorking      ci-cd-project   githubaction-project  merge_conflict_demo  revert_demo     student-repo
root@LAPTOP-HPUOTJF1:~# cd httpd/
root@LAPTOP-HPUOTJF1:~/httpd# ls
2.4  LICENSE  README.md  generate-stackbrew-library.sh  update.sh
root@LAPTOP-HPUOTJF1:~/httpd# docker build -t httpd-custom:latest -f 2.4/Dockerfile .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 388.6kB
Step 1/14 : FROM debian:trixie-slim
trixie-slim: Pulling from library/debian
8ff677ca96a9: Download complete
Digest: sha256:020c0d20b988058cbe785a9db107156c3c75c2ac944a6aa7ab59f2add76a7bd
Status: Downloaded newer image for debian:trixie-slim
--> 020c0d20b988
Step 2/14 : ENV HTTPD_PREFIX /usr/local/apache2
--> Running in 93137df01e72
--> Removed intermediate container 93137df01e72
--> 2d237400f95a
Step 3/14 : ENV PATH $HTTPD_PREFIX/bin:$PATH
--> Running in 4564cb53df62
--> Removed intermediate container 4564cb53df62
--> d153592a9cd0
Step 4/14 : RUN mkdir -p "$HTTPD_PREFIX" && chown www-data:www-data "$HTTPD_PREFIX"
--> Running in a0eee706afa9
--> Removed intermediate container a0eee706afa9
--> 99646c12e01b
Step 5/14 : WORKDIR $HTTPD_PREFIX

COPY failed: file not found in build context or excluded by .dockerignore: stat httpd-foreground: file does not exist
root@LAPTOP-HPUOTJF1:~/httpd# docker images

INFO In Use
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
achalpadol/ubuntu:latest            3131b4cc82a7                      161MB       45.3MB       U
achalpadol/ubuntu:v1                3131b4cc82a7                      161MB       45.3MB       U
achalpadol_redis:version1           234c902a2db4                      208MB       56.3MB       U
achalpadol_redis:version3           234c902a2db4                      208MB       56.3MB       U
debian:trixie-slim                  020c0d20b988                      119MB       31.8MB       U
httpd:latest                         305fd8326a27                      177MB       47.6MB       U
lancachenet/ubuntu:latest           f86a1b63836e                      248MB       61.4MB       U
nginx:latest                        b5a9a3cfe86b                      241MB       66MB         U
redis:latest                        234c902a2db4                      208MB       56.3MB       U
ubuntu-nginx:latest                 5102563991c6                      315MB       80.4MB       U
ubuntu:latest                       3131b4cc82a7                      161MB       45.3MB       U
root@LAPTOP-HPUOTJF1:~/httpd# docker run -d --name apache-web -p 8181:80 httpd-custom:latest
Unable to find image 'httpd-custom:latest' locally
docker: Error response from daemon: pull access denied for httpd-custom, repository does not exist or may require 'docker login'

Run 'docker run --help' for more information
root@LAPTOP-HPUOTJF1:~/httpd# docker run -d --name apache-web -p 8181:80 httpd:latest
6ad5ea0758015fc2dad2bda5107ef2e807c01ffb0df45d6c6fca85c4d74b92ae
root@LAPTOP-HPUOTJF1:~/httpd# docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS                    NAMES
6ad5ea075801   httpd:latest                        "httpd-foreground"      13 seconds ago Up 12 seconds  0.0.0.0:8181->80/tcp, [::]:8181->80/tcp  apache-web
6ad5ea075801   achalpadol_redis:version3           "docker-entrypoint.s"   10 minutes ago Up 10 minutes  6379/tcp               redis03
```

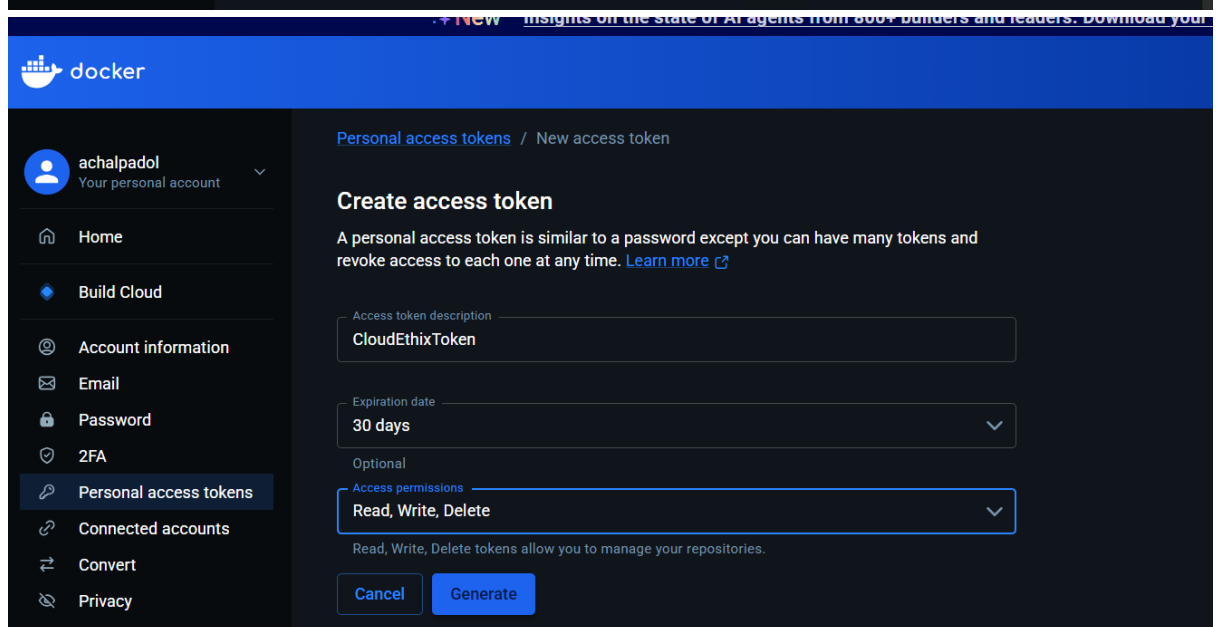
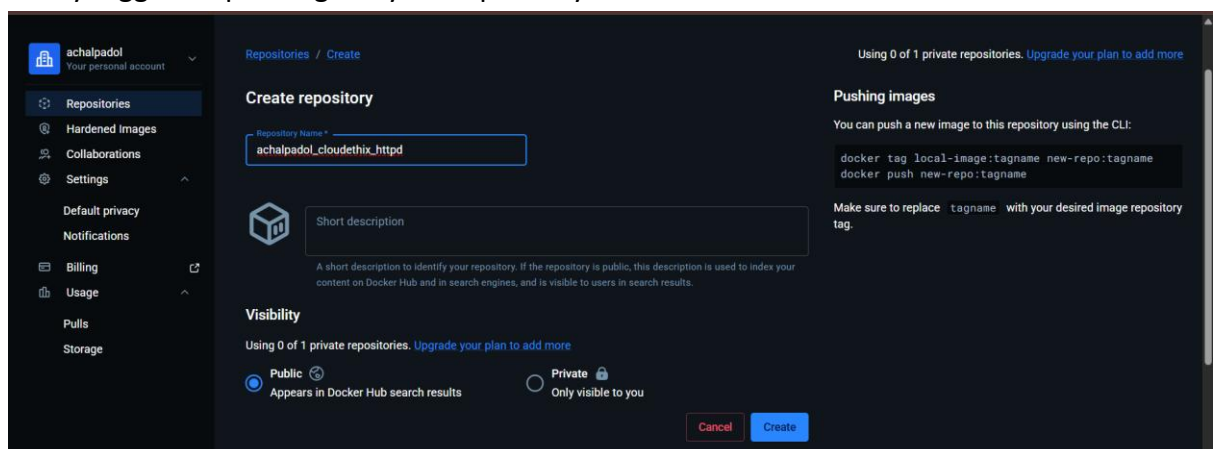
It works!

- tag the image as V1.

```
977d5e5c948f  httpd  "httpd-foreground"  38 minutes ago
root@LAPTOP-HPU0TJF1:~/httpd# docker tag httpd:latest httpd-custom:V1
root@LAPTOP-HPU0TJF1:~/httpd# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
achalpadol/ubuntu:latest	3131b4cc82a7	161MB	45.3MB	U
achalpadol/ubuntu:v1	3131b4cc82a7	161MB	45.3MB	U
achalpadol_redis:version1	234c902a2db4	208MB	56.3MB	U
achalpadol_redis:version3	234c902a2db4	208MB	56.3MB	U
debian:trixie-slim	020c0d20b988	119MB	31.8MB	
httpd-custom:V1	305fd8326a27	177MB	47.6MB	U

22. Create a Docker Hub account and generate a security token. Establish a Docker Hub repository named "yourname_cloudethix_httpd." Log in to Docker Hub and push the previously tagged httpd image to your repository.




```

root@LAPTOP-HPUOTJF1:~# docker tag achalpadol_redis:version1 achalpadol/achalpadol_cloudethix_redis:version1
root@LAPTOP-HPUOTJF1:~# docker tag achalpadol_redis:version3 achalpadol/achalpadol_cloudethix_redis:version3
root@LAPTOP-HPUOTJF1:~# docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
achalpadol/achalpadol_cloudethix_httpd:V1    305fd8326a27    177MB      47.6MB      U
achalpadol/achalpadol_cloudethix_redis:version1 234c902a2db4    208MB      56.3MB      U
achalpadol/achalpadol_cloudethix_redis:version3 234c902a2db4    208MB      56.3MB      U
achalpadol/ubuntu:latest                    3131b4cc82a7    161MB      45.3MB      U

root@LAPTOP-HPUOTJF1:~# docker tag achalpadol_redis:version1 achalpadol/achalpadol_cloudethix_redis:version1
root@LAPTOP-HPUOTJF1:~# docker tag achalpadol_redis:version3 achalpadol/achalpadol_cloudethix_redis:version3
root@LAPTOP-HPUOTJF1:~# docker images

IMAGE                                ID                                DISK USAGE  CONTENT SIZE  EXTRA
achalpadol/achalpadol_cloudethix_httpd:V1    305fd8326a27    177MB      47.6MB      U
achalpadol/achalpadol_cloudethix_redis:version1 234c902a2db4    208MB      56.3MB      U
achalpadol/achalpadol_cloudethix_redis:version3 234c902a2db4    208MB      56.3MB      U
achalpadol/ubuntu:latest                    3131b4cc82a7    161MB      45.3MB      U
achalpadol/ubuntu:V1                        3131b4cc82a7    161MB      45.3MB      U
achalpadol_redis:version1                   234c902a2db4    208MB      56.3MB      U
achalpadol_redis:version3                   234c902a2db4    208MB      56.3MB      U
debian:trixie-slim                          020c9d20b98b    119MB      31.8MB      U
httpd-custom:V1                             305fd8326a27    177MB      47.6MB      U
httpd:latest                                305fd8326a27    177MB      47.6MB      U
lancachenet/ubuntu:latest                   f86a1b63836e    248MB      61.4MB      U
nginx:latest                                b5a9a3cfc86b    241MB      66MB        U
redis:latest                                234c902a2db4    208MB      56.3MB      U
ubuntu-nginx:latest                         3102563991c6    315MB      80.4MB      U
ubuntu:latest                               3131b4cc82a7    161MB      45.3MB      U

root@LAPTOP-HPUOTJF1:~# docker push achalpadol/achalpadol_cloudethix_redis:version1
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_redis]
062e456697fa: Mounted from achalpadol/achalpadol_cloudethix_httpd
fd6b3883006b: Mounted from library/redis
6f0ec8fc933c: Mounted from library/redis
c1af3958d6f8: Mounted from library/redis
234dd6735104: Mounted from library/redis
8fd80e87c6f1: Mounted from library/redis
4f4fb700ef54: Mounted from achalpadol/achalpadol_cloudethix_httpd
version1: digest: sha256:5d2c689b4b55fc3fab4b0cc8aaa950f85b508c76cle0f35a90d8f411d55a8b2b size: 2286

Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:234c902a2db40461a129e2d4aeff85b28cf20187ed274ac7fee50995fa713c7b -> sha256:5d2c689b4b55fc3fab4b0cc8aaa950f85b508c76cle0f35a90d8f411d55a8b2b

root@LAPTOP-HPUOTJF1:~# docker push achalpadol/achalpadol_cloudethix_redis:version3
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_redis]
4f4fb700ef54: Layer already exists
fd6b3883006b: Layer already exists
234dd6735104: Layer already exists
062e456697fa: Layer already exists
c1af3958d6f8: Layer already exists
8fd80e87c6f1: Layer already exists
6f0ec8fc933c: Layer already exists
version3: digest: sha256:5d2c689b4b55fc3fab4b0cc8aaa950f85b508c76cle0f35a90d8f411d55a8b2b size: 2286

Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:234c902a2db40461a129e2d4aeff85b28cf20187ed274ac7fee50995fa713c7b -> sha256:5d2c689b4b55fc3fab4b0cc8aaa950f85b508c76cle0f35a90d8f411d55a8b2b

root@LAPTOP-HPUOTJF1:~# docker push achalpadol/achalpadol_cloudethix_redis:version3
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_redis]
4f4fb700ef54: Layer already exists
fd6b3883006b: Layer already exists
234dd6735104: Layer already exists
062e456697fa: Layer already exists
c1af3958d6f8: Layer already exists
8fd80e87c6f1: Layer already exists
6f0ec8fc933c: Layer already exists
version3: digest: sha256:5d2c689b4b55fc3fab4b0cc8aaa950f85b508c76cle0f35a90d8f411d55a8b2b size: 2286

Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:234c902a2db40461a129e2d4aeff85b28cf20187ed274ac7fee50995fa713c7b -> sha256:5d2c689b4b55fc3fab4b0cc8aaa950f85b508c76cle0f35a90d8f411d55a8b2b

```

24. Create a remote Git repository and add the Dockerfile and index.html files. Clone the repository locally and create a release branch. Build the Docker image from the release branch with a meaningful tag, then run a container from the image and expose it on host port 8383. Check the webpage in a browser, and upon success, push the image to your

Docker Hub repository named "yourname_cloudethix_nginx." Finally, push the release branch to the remote Git repository and merge it by creating a pull request.

```
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# git clone https://github.com/achalpadol/cloudethix-nginx.git
Cloning into 'cloudethix-nginx'...
remote: Enumerating objects: 6, done.
remote: Counting objects: 100% (6/6), done.
remote: Compressing objects: 100% (5/5), done.
remote: Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (6/6), done.
root@LAPTOP-HPUOTJF1:~# ls
1ff                                DevOps_Project  Merge_demo      clone-remote-repo  githubaction-project  merge_confilct_demo  revert_demo      student-repo
Amend_repo                        GitIgnore-Repo  amend-repo      cloudethix-nginx   httpd                 project             stash-demo       ubuntu-nginx
Cloud_Ethix_Copyfiles            GitWorking      ci-cd-project   free-for-dev       index.html             revert-demo          stash1-demo
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# ls
Dockerfile  index.html
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git checkout -b release
Switched to a new branch 'release'
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git brachn
git: 'brachn' is not a git command. See 'git --help'.

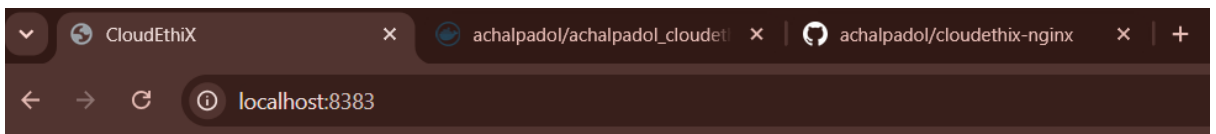
The most similar command is
branch
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git branch
main
* release
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker build -t nginx-release: v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 66.56kB
Step 1/3 : FROM nginx:latest
----> b5a9a3cfc86b
Step 2/3 : COPY index.html /usr/share/nginx/html/index.html
----> cc17b8dbb198
Step 3/3 : EXPOSE 80
----> Running in efe48c186468
----> Removed intermediate container efe48c186468
----> e931316b8da7
Successfully built e931316b8da7
Successfully tagged nginx-release:v1
```

run a container from the image and expose it on host port 8383.

```
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker run -d --name nginx-release -p 8383:80 nginx-release:v1
7d6c8bf84ddb319f7f8f57bd5a387a6d8c7444fad11397e737ba0f3e3bd5e0b
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
7d6c8bf84ddb   nginx-release:v1  "/docker-entrypoint..."  3 seconds ago  Up 3 seconds  0.0.0.0:8383->80/tcp, [::]:8383->80/tcp  nginx-release
6ad5ea075881   httpd:latest    "httpd-foreground"        44 minutes ago  Up 44 minutes  0.0.0.0:8181->80/tcp, [::]:8181->80/tcp  apache-web
```

. Check the webpage in a browser



Hello there, Let's be the Team CloudEthiX

push the image to your Docker Hub repository named

```
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker push achalpadol/achalpadol_cloudethix_nginx:v1
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_nginx]
39694cde734e: Mounted from library/nginx
45ef33eb91c8: Mounted from library/nginx
8d2092eabbc5: Mounted from library/nginx
ace9659054fc: Pushed
0b19be717994: Mounted from library/nginx
3b9f33509a826: Mounted from library/nginx
062e450697fa: Mounted from achalpadol/achalpadol_cloudethix_redis
6a24a6ebcf30: Mounted from library/nginx
v1: digest: sha256:e931316b8da71ccc4620e346cd5b3aabe715870595105fe7547bb860041a30f size: 2043
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx#
```

Finally, push the release branch to the remote Git repository and merge it by creating a pull request.

```

root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git push -u origin release
Username for 'https://github.com': achalpadol
Password for 'https://achalpadol@github.com':
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'release' on GitHub by visiting:
remote:      https://github.com/achalpadol/cloudethix-nginx/pull/new/release
remote:
To https://github.com/achalpadol/cloudethix-nginx.git
 * [new branch]      release -> release
branch 'release' set up to track 'origin/release'.
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# |

```

25. Save all local Redis images as a .tar file in the master branch of your local repository. Delete all Redis images from your local system and push the master branch to the remote repository. Load the Redis images from the tar file to your local system, and verify that all images have been loaded correctly.

```

Switched to branch 'master'
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker save -o redis-images.tar \
achalpadol_redis:version1 \
achalpadol_redis:version3
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# ls
Dockerfile  index.html  redis-images.tar
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git add redis-images.tar
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git commit -m "Added Redis Docker images tar file"
[master 8631763] Added Redis Docker images tar file
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 redis-images.tar
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# git push origin master
Username for 'https://github.com': achalpadol
Password for 'https://achalpadol@github.com':
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 12 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (3/3), 51.66 MiB | 914.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: warning: See https://gh.io/lfs for more information.
remote: warning: File redis-images.tar is 53.71 MB; this is larger than GitHub's recommended maximum file size of 50.00 MB
remote: warning: GH001: Large files detected. You may want to try Git Large File Storage - https://git-lfs.github.com.
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/achalpadol/cloudethix-nginx/pull/new/master
remote:
To https://github.com/achalpadol/cloudethix-nginx.git
 * [new branch]      master -> master
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker rmi achalpadol_redis:version1
Untagged: achalpadol_redis:version1
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker rmi achalpadol_redis:version3
Untagged: achalpadol_redis:version3
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
achalpadol/achalpadol_cloudethix_httpd:v1	305fd8326a27	177MB	47.6MB	U
achalpadol/achalpadol_cloudethix_nginx:v1	e931316b8da7	238MB	63.1MB	U

- . Load the Redis images from the tar file to your local system, and verify that all images have been loaded correctly.

```

root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker load -i redis-images.tar
Loaded image: achalpadol_redis:version1
Loaded image: achalpadol_redis:version3
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# docker images

```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
achalpadol/achalpadol_cloudethix_httpd:v1	305fd8326a27	177MB	47.6MB	U
achalpadol/achalpadol_cloudethix_nginx:v1	e931316b8da7	238MB	63.1MB	U
achalpadol/achalpadol_cloudethix_redis:version1	234c902a2db4	208MB	56.3MB	U
achalpadol/achalpadol_cloudethix_redis:version3	234c902a2db4	208MB	56.3MB	U
achalpadol/ubuntu:latest	3131b4cc82a7	161MB	45.3MB	U
achalpadol/ubuntu:v1	3131b4cc82a7	161MB	45.3MB	U
achalpadol_redis:version1	234c902a2db4	208MB	56.3MB	U
achalpadol_redis:version3	234c902a2db4	208MB	56.3MB	U
debian:trixie-slim	020c0d20b988	119MB	31.8MB	

26. Pull the Busy-box image to your local system, tag it, and push it to the Docker Hub repository "yourname_cloudethix_busybox." Export the Docker image from the NGINX container, create a .tar file, and import the tar file to create a Docker image with a meaningful name. After importing the image, tag it and push it to the "yourname_cloudethix_busybox" Docker Hub repository.

- Pull the busy-box image to your local system

```
root@LAPTOP-HPUOTJF1:~/cloudethix-nginx# cd
root@LAPTOP-HPUOTJF1:~/# docker pull busybox
Using default tag: latest
latest: Pulling from library/busybox
b05093807bb0: Pull complete
7270b3e1860c: Download complete
Digest: sha256:fd8d9aa63ba2f0982b5304e1ee8d3b90a210bc1ffb5314d980eb6962f1a9715d
Status: Downloaded newer image for busybox:latest
docker.io/library/busybox:latest
root@LAPTOP-HPUOTJF1:~/# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
achalpadol/achalpadol_cloudethix_httpd:v1	305fd8326a27	177MB	47.6MB	U
achalpadol/achalpadol_cloudethix_nginx:v1	e931316b8da7	238MB	63.1MB	U
achalpadol/achalpadol_cloudethix_redis:version1	234c902a2db4	208MB	56.3MB	U
achalpadol/achalpadol_cloudethix_redis:version3	234c902a2db4	208MB	56.3MB	U
achalpadol/ubuntu:latest	3131b4cc82a7	161MB	45.3MB	U
achalpadol/ubuntu:v1	3131b4cc82a7	161MB	45.3MB	U
achalpadol_redis:version1	234c902a2db4	208MB	56.3MB	U
achalpadol_redis:version3	234c902a2db4	208MB	56.3MB	U
busybox:latest	fd8d9aa63ba2	6.81MB	2.23MB	

- tag it and push to the Docker Hub

```
ubuntu:nginx:latest 3131b4cc82a7 161MB 45.3MB U
ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
root@LAPTOP-HPUOTJF1:~/# docker push achalpadol/achalpadol_cloudethix_busybox:v1
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_busybox]
b05093807bb0: Mounted from library/busybox
v1: digest: sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8 size: 610

Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:fd8d9aa63ba2f0982b5304e1ee8d3b90a210bc1ffb5314d980eb6962f1a9715d -> sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8
root@LAPTOP-HPUOTJF1:~/# |
```

- Export the Docker image from the NGINX container, create a .tar file

```
root@LAPTOP-HPUOTJF1:~/# docker ps
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
7dc68bf84dbb nginx-release:v1 "/docker-entrypoint..." 46 minutes ago Up 46 minutes 0.0.0.0:8383->80/tcp, [::]:8383->80/tcp nginx-release
6ad5ea075801 httpd:latest "httpd-foreground" 2 hours ago Up 2 hours 0.0.0.0:8181->80/tcp, [::]:8181->80/tcp apache-web
ab6aaf3634f0 achalpadol_redis:version3 "docker-entrypoint.s..." 2 hours ago Up 2 hours 6379/tcp redis02
b37b5a6ff268 achalpadol_redis:version1 "docker-entrypoint.s..." 2 hours ago Up 2 hours 6379/tcp redis01
1b33c35e955b nginx "httpd-foreground" 2 hours ago Up 2 hours 0.0.0.0:32768->80/tcp, [::]:32768->80/tcp nginx-web
977d5e5c948f httpd "httpd-foreground" 2 hours ago Up 2 hours 0.0.0.0:32768->80/tcp, [::]:32768->80/tcp httpd-always

root@LAPTOP-HPUOTJF1:~/# docker export -o nginx-container.tar nginx-web
root@LAPTOP-HPUOTJF1:~/# ls
lff DevOps_Project Merge_demo clone-remote-repo githubaction-project merge_conflict_demo revert-demo stash1-demo
Amend_repo GitIgnore-Repo amend-repo cloudethix-nginx httpd nginx-container.tar nginx-container.tar revert-demo student-repo
Cloud_Ethix_Copyfiles GitWorking ci-cd-project free-for-dev index.html project project stash-demo ubuntu-nginx
```

- import the tar file to create a Docker image with a meaningful name.

```
Amend_repo GitIgnore-Repo amend-repo cloudethix-nginx httpd nginx-container.tar nginx-container.tar revert-demo student-repo
Cloud_Ethix_Copyfiles GitWorking ci-cd-project free-for-dev index.html project project stash-demo ubuntu-nginx
root@LAPTOP-HPUOTJF1:~/# docker import nginx-container.tar nginx-imported:v1
sha256:ce7e04831e7acde10f06ba4a4807fd9524e299121534a9999060c92f5826585
root@LAPTOP-HPUOTJF1:~/# docker tag nginx-imported:v1 achalpadol/achalpadol_cloudethix_busybox:nginx-v1
root@LAPTOP-HPUOTJF1:~/# docker push achalpadol/achalpadol_cloudethix_busybox:nginx-v1
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_busybox]
23011aeeb495: Pushed
nginx-v1: digest: sha256:ce7e04831e7acde10f06ba4a4807fd9524e299121534a9999060c92f5826585 size: 406
root@LAPTOP-HPUOTJF1:~/# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
achalpadol/achalpadol_cloudethix_busybox:v1	fd8d9aa63ba2	6.81MB	2.23MB	
achalpadol/achalpadol_cloudethix_busybox:nginx-v1	ce7e04831e7a	236MB	63.7MB	
achalpadol/achalpadol_cloudethix_httpd:v1	305fd8326a27	177MB	47.6MB	U
achalpadol/achalpadol_cloudethix_nginx:v1	e931316b8da7	238MB	63.1MB	U
achalpadol/achalpadol_cloudethix_redis:version1	234c902a2db4	208MB	56.3MB	U

- push it to the "yourname_cloudethix_busybox" Docker Hub repository.

```
ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
root@LAPTOP-HPUOTJF1:~/# docker push achalpadol/achalpadol_cloudethix_busybox:v1
The push refers to repository [docker.io/achalpadol/achalpadol_cloudethix_busybox]
b05093807bb0: Mounted from library/busybox
v1: digest: sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8 size: 610

Info → Not all multiplatform-content is present and only the available single-platform image was pushed
sha256:fd8d9aa63ba2f0982b5304e1ee8d3b90a210bc1ffb5314d980eb6962f1a9715d -> sha256:1cfa4e2b09e127b9c4ed43578d3f3c18e7d44ea47b9ea98475c0cbe9086525f8
root@LAPTOP-HPUOTJF1:~/# docker ps
```

27. **Dockerfile creation:** Create a simple Dockerfile to build a custom image with the following specifications:

- Base image: Ubuntu
- Install packages: curl, vim, and git
- Set an environment variable: MY_NAME=Your_Name

-Build the custom image using docker build and list all available images using docker images.

```
FROM ubuntu:latest

LABEL maintainer="Achal Padol"

RUN apt-get update && \
    apt-get install -y curl vim git && \
    apt-get clean && \
    rm -rf /var/lib/apt/lists/*

ENV MY_NAME="Achal Padol"

CMD ["/bin/bash"]
~
~
~
~
~
~
~

ubuntu:latest 3131b4cc82a7 161MB 45.3MB U
root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# mkdir ubuntu-custom
root@LAPTOP-HPUOTJF1:~# cd ubuntu-custom/
root@LAPTOP-HPUOTJF1:~/ubuntu-custom# vim Dockerfile
root@LAPTOP-HPUOTJF1:~/ubuntu-custom# vim Dockerfile
root@LAPTOP-HPUOTJF1:~/ubuntu-custom# docker build -t ubuntu-custom:v1 .
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
Install the buildx component to build images with BuildKit:
https://docs.docker.com/go/buildx/

Sending build context to Docker daemon 2.048kB
Step 1/5 : FROM ubuntu:latest
----> 3131b4cc82a7
Step 2/5 : LABEL maintainer="Achal Padol"
----> Running in f0301f9f134e
----> Removed intermediate container f0301f9f134e
----> 8ba7de75c849
Step 3/5 : RUN apt-get update && apt-get install -y curl vim git && apt-get clean && rm -rf /var/lib/apt/lists/*
Successfully built 5af056060ff5
Successfully tagged ubuntu-custom:v1
root@LAPTOP-HPUOTJF1:~/ubuntu-custom# docker images
```

IMAGE	ID	DISK USAGE	CONTENT SIZE	EXTRA
achalpadol/achalpadol_cloudethix_busybox:V1	fd8d9aa63ba2	6.81MB	2.23MB	
achalpadol/achalpadol_cloudethix_busybox:nginx-v1	ce7e04831e7a	236MB	63.7MB	
achalpadol/achalpadol_cloudethix_httpd:V1	305fd8326a27	177MB	47.6MB	U
achalpadol/achalpadol_cloudethix_nginx:v1	e931316b8da7	238MB	63.1MB	U
achalpadol/achalpadol_cloudethix_redis:version1	234c902a2db4	208MB	56.3MB	U
achalpadol/achalpadol_cloudethix_redis:version3	234c902a2db4	208MB	56.3MB	U
achalpadol/ubuntu:latest	3131b4cc82a7	161MB	45.3MB	U
achalpadol/ubuntu:v1	3131b4cc82a7	161MB	45.3MB	U
achalpadol_redis:version1	234c902a2db4	208MB	56.3MB	U
achalpadol_redis:version3	234c902a2db4	208MB	56.3MB	U
busybox:latest	fd8d9aa63ba2	6.81MB	2.23MB	
debian:trixie-slim	020c0d20b988	119MB	31.8MB	
httpd-custom:V1	305fd8326a27	177MB	47.6MB	U
httpd:latest	305fd8326a27	177MB	47.6MB	U
lancachenet/ubuntu:latest	f86a1b63836e	248MB	61.4MB	
nginx-imported:v1	ce7e04831e7a	236MB	63.7MB	
nginx-release:v1	e931316b8da7	238MB	63.1MB	U
nginx:latest	b5a9a3cfc86b	241MB	66MB	U
redis:latest	234c902a2db4	208MB	56.3MB	U
ubuntu-custom:v1	5af056060ff5	407MB	98.9MB	

28. **Docker networking:** Create two Docker containers with different names, and add them to the same custom network. Verify that the containers can communicate with each other using their names as hostnames.

-Create Docker Network

```

root@LAPTOP-HPUOTJF1:~/ubuntu-custom# cd
root@LAPTOP-HPUOTJF1:~# docker network create cloud-network
a964394174fa54a200110e17d82484e7b10a72c6b8ffd3f98544bcea482fda69
root@LAPTOP-HPUOTJF1:~# docker network ls

```

NETWORK ID	NAME	DRIVER	SCOPE
14516bb6eaf3	bridge	bridge	local
a964394174fa	cloud-network	bridge	local

-Create two Docker containers with different names, and add them to the same custom network.

```

root@LAPTOP-HPUOTJF1:~# docker run -dit --name ubuntu01 --network cloud-network ubuntu bash
9b93c1eae8897159c488110ed58ca63759afc484d1778ce3b6e960d180e2455e
root@LAPTOP-HPUOTJF1:~# docker run -dit --name ubuntu02 --network cloud-network ubuntu bash
d7f26af5dcf3997fde721ea93335ff6c1a789778c5a557d62038a73605bb3d8b
root@LAPTOP-HPUOTJF1:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
d7f26af5dcf3	ubuntu	"bash"	5 seconds ago	Up 4 seconds		ubuntu02
9b93c1eae889	ubuntu	"bash"	19 seconds ago	Up 18 seconds		ubuntu01

```

root@9b93c1eae889:/# exit
exit
root@LAPTOP-HPUOTJF1:~# docker exec -it ubuntu01 bash
root@9b93c1eae889:/# ping ubuntu02
PING ubuntu02 (172.20.0.3) 56(84) bytes of data:
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=1 ttl=64 time=0.543 ms
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=2 ttl=64 time=0.302 ms
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=3 ttl=64 time=0.195 ms
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=4 ttl=64 time=0.132 ms
64 bytes from ubuntu02.cloud-network (172.20.0.3): icmp_seq=5 ttl=64 time=0.068 ms

```

29. **Data persistence with volumes:** Run a Docker container with an attached volume, create a file inside the volume, and verify that the file persists after the container is stopped and removed.

-Create a container with attach volume

```

bash: docker: command not found
root@9b93c1eae889:/# docker volume create cloud-data
bash: docker: command not found
root@9b93c1eae889:/# exit
exit
root@LAPTOP-HPUOTJF1:~# docker run -dit --name ubuntu-volume -v cloud-data:/data ubuntu bash
e43fbb4322545b75518db089111ded2abcfb86ec80f5dfd27050c6f5620d81a8
root@LAPTOP-HPUOTJF1:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e43fbb432254	ubuntu	"bash"	14 seconds ago	Up 14 seconds		ubuntu-volume
d7f26af5dcf3	ubuntu	"bash"	11 minutes ago	Up 11 minutes		ubuntu02
9b93c1eae889	ubuntu	"bash"	11 minutes ago	Up 11 minutes		ubuntu01

-Create a file inside the volume

```

root@LAPTOP-HPUOTJF1:~# docker exec -it ubuntu-volume bash
root@e43fbb432254:/# echo "Hello CloudEthiX" > /data/file.txt
root@e43fbb432254:/# cat /data/file.txt
Hello CloudEthiX
root@e43fbb432254:/# exit
exit
root@LAPTOP-HPUOTJF1:~#

```

- verify that the file persists after the container is stopped and removed.

```

Run "docker volume COMMAND --help" for more information on a command.
root@LAPTOP-HPUOTJF1:~# docker volume ls

```

DRIVER	VOLUME NAME
local	cab24f1de152a526001218ea642fd77196d83e8780175aff2e52fe311dd2696b
local	cloud-data

```

root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# docker rm -f ubuntu-volume
ubuntu-volume
root@LAPTOP-HPUOTJF1:~#

```

-create new container and attach volume to the new container and verify the data

```

root@LAPTOP-HPUOTJF1:~# docker rm -f ubuntu-volume
ubuntu-volume
root@LAPTOP-HPUOTJF1:~# docker run -dit --name ubuntu-volume-new \
-v cloud-data:/data \
ubuntu bash
c67f6cb39973ce71ec4f8801a54fc412fc048c18ec8927441036599afb4c3517
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
c67f6cb39973   ubuntu    "bash"                  4 seconds ago Up 3 seconds                                     ubuntu-volume-new
d9f26af5dcf3   ubuntu    "bash"                  17 minutes ago Up 17 minutes                                     ubuntu02
9b93c1eae889   ubuntu    "bash"                  18 minutes ago Up 18 minutes                                     ubuntu01
7d6c8bf84dbb   nginx-release:vl "/docker-entrypoint.s..." 2 hours ago   Up 2 hours    0.0.0.0:8383->80/tcp, [::]:8383->80/tcp nginx-release
6ad5ea075801   httpd:latest "httpd-foreground"      2 hours ago   Up 2 hours    0.0.0.0:8181->80/tcp, [::]:8181->80/tcp apache-web
ab6aaf3634f0   achalpadol_redis:version3 "docker-entrypoint.s..." 2 hours ago   Up 2 hours    6379/tcp                               redis02

```

```

root@LAPTOP-HPUOTJF1:~#
docker exec -it ubuntu-volume-new bash
root@c67f6cb39973:/# ls /data
file.txt
root@c67f6cb39973:/# cat /data/file.txt
Hello CloudEthiX
root@c67f6cb39973:/# |

```

30. **Docker Compose:** Create a docker-compose.yml file to run a multi-container application consisting of a web server (e.g., Nginx) and a database server (e.g., MySQL). Ensure that both containers are connected to the same network.

-create docker compose file to create multi-container

```

version: "3.8"

services:
  web:
    image: nginx:latest
    container_name: nginx-web
    ports:
      - "8080:80"
    networks:
      - cloud-network

  db:
    image: mysql:8.0
    container_name: mysql-db
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: cloudethix
      MYSQL_USER: achal
      MYSQL_PASSWORD: password123
    ports:
      - "3306:3306"
    networks:
      - cloud-network

networks:
  cloud-network:
    driver: bridge

```

-run both the container

```

root@LAPTOP-HPUOTJF1:~/docker-compose-demo# cd
root@LAPTOP-HPUOTJF1:~/docker-compose-demo# vim docker-compose.yml
root@LAPTOP-HPUOTJF1:~/docker-compose-demo# docker compose up -d
WARN[0000] /root/docker-compose-demo/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 2/2
  ✓ Container nginx-web  Running
  ✓ Container mysql-db   Started
root@LAPTOP-HPUOTJF1:~/docker-compose-demo# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
cf35c688c237   mysql:8.0   "docker-entrypoint.s..." 12 seconds ago Up 12 seconds    3306/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp mysql-db
9e9d68ee8027   nginx:latest "/docker-entrypoint.s..." 3 minutes ago  Up 3 minutes    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp nginx-web
c67f6cb39973   ubuntu    "bash"                  14 minutes ago Up 14 minutes                                     ubuntu-volum

```

- Ensure that both containers are connected to the same network.

```

    "com.docker.compose.network": "cloud-network",
    "com.docker.compose.project": "docker-compose-demo",
    "com.docker.compose.version": "2.40.3"
  },
  "Containers": {
    "9e9d68ee8027b762e315507a633f8dbf97fa6ee8c7257911f8aa6d6595e76649": {
      "Name": "nginx-web",
      "EndpointID": "17f1236ba6dc74bbda814b9aea25a5aa0ee31536490169522113bb9f152e62fb",
      "MacAddress": "7a:ce:29:2e:f5:de",
      "IPv4Address": "172.21.0.2/16",
      "IPv6Address": ""
    },
    "cf35c688c23712aa8bc4e8460b3a1a181b5d698e4cacdbc16d08283f07dcb174": {
      "Name": "mysql-db",
      "EndpointID": "2c92904f3790cef5da502e1e7a6f429aada6d2b778560dc3ea4cb0c78c076e61",
      "MacAddress": "5e:1f:09:5c:64:71",
      "IPv4Address": "172.21.0.3/16",
      "IPv6Address": ""
    }
  }
}

```

31. **Container logging:** Run a Docker container that generates log output, and practice using docker logs to view and analyze the logs.

```

root@LAPTOP-HPUOTJF1:~# docker run -d --name nginx-logs -p 8081:80 nginx
8f30dbddbbf1271588b555dfaf6345de056fc37e810ad59709168cca9a8bdbaf
root@LAPTOP-HPUOTJF1:~# docker ps

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
8f30dbddbbf1	nginx	"/docker-entrypoint..."	6 seconds ago	Up 5 seconds	0.0.0.0:8081->80/tcp, [::]:8081->80/tcp	nginx-logs

View logs

```

root@LAPTOP-HPUOTJF1:~# docker logs nginx-logs
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/07/22 06:40:26 [notice] 1#1: using the "epoll" event method
2026/07/22 06:40:26 [notice] 1#1: nginx/1.31.2
2026/07/22 06:40:26 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/07/22 06:40:26 [notice] 1#1: OS: Linux 6.6.114.1-microsoft-standard-WSL2
2026/07/22 06:40:26 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1024:1048576
2026/07/22 06:40:26 [notice] 1#1: start worker processes
2026/07/22 06:40:26 [notice] 1#1: start worker process 29
2026/07/22 06:40:26 [notice] 1#1: start worker process 30
2026/07/22 06:40:26 [notice] 1#1: start worker process 31
2026/07/22 06:40:26 [notice] 1#1: start worker process 32
2026/07/22 06:40:26 [notice] 1#1: start worker process 33
2026/07/22 06:40:26 [notice] 1#1: start worker process 34
2026/07/22 06:40:26 [notice] 1#1: start worker process 35
2026/07/22 06:40:26 [notice] 1#1: start worker process 36
2026/07/22 06:40:26 [notice] 1#1: start worker process 37
2026/07/22 06:40:26 [notice] 1#1: start worker process 38
2026/07/22 06:40:26 [notice] 1#1: start worker process 39
2026/07/22 06:40:26 [notice] 1#1: start worker process 40
172.17.0.1 - - [22/Jul/2026:06:40:55 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:40:59 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:01 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:02 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:03 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:05 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
root@LAPTOP-HPUOTJF1:~# |
172.17.0.1 - - [22/Jul/2026:06:41:05 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
^Croot@LAPTOP-HPUOTJF1:~# docker logs --tail 5 nginx-logs
172.17.0.1 - - [22/Jul/2026:06:40:59 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:01 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:02 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:03 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
172.17.0.1 - - [22/Jul/2026:06:41:05 +0000] "GET / HTTP/1.1" 200 896 "-" "curl/8.18.0" "-"
root@LAPTOP-HPUOTJF1:~# |

```

32. **Container resource monitoring:** Run a Docker container that consumes system resources (e.g., CPU or memory) and practice using docker stats and docker top to monitor the container's resource usage.

-docker stats


```

root@LAPTOP-HPUOTJF1:~/nginx-update# docker build -t nginx-app:v1
DEPRECATED: The legacy builder is deprecated and will be removed in a future release.
            Install the buildx component to build images with BuildKit:
            https://docs.docker.com/go/buildx/

Sending build context to Docker daemon  3.072kB
Step 1/3 : FROM nginx:latest
----> b5a9a3cfc86b
Step 2/3 : COPY index.html /usr/share/nginx/html/index.html
----> 52aad059802
Step 3/3 : EXPOSE 80
----> Running in 36f8efd2c144
----> Removed intermediate container 36f8efd2c144
----> 9d124c9d5add
Successfully built 9d124c9d5add
Successfully tagged nginx-app:v1
root@LAPTOP-HPUOTJF1:~/nginx-update# docker images

IMAGE                                                    ID                                DISK USAGE    CONTENT SIZE    EXTRA
achalpadol/achalpadol_cloudethix_busybox:V1            fd8d9aa63ba2                      6.81MB         2.23MB
achalpadol/achalpadol_cloudethix_busybox:nginx-v1      ce7e04831e7a                      236MB          63.7MB
achalpadol/achalpadol_cloudethix_httpd:V1              305fd8326a27                      177MB           47.6MB
achalpadol/achalpadol_cloudethix_nginx:v1              e931316b8da7                      238MB           63.1MB
achalpadol/achalpadol_cloudethix_redis:version1        234c902a2db4                      208MB           56.3MB
achalpadol/achalpadol_cloudethix_redis:version3        234c902a2db4                      208MB           56.3MB
achalpadol/ubuntu:latest                               3131b4cc82a7                      161MB           45.3MB
achalpadol/ubuntu:v1                                   3131b4cc82a7                      161MB           45.3MB
achalpadol_redis:version1                               234c902a2db4                      208MB           56.3MB
achalpadol_redis:version3                               234c902a2db4                      208MB           56.3MB
busybox:latest                                           fd8d9aa63ba2                      6.81MB         2.23MB
debian:trixie-slim                                      020c0d20b988                      119MB           31.8MB
httpd-custom:V1                                          305fd8326a27                      177MB           47.6MB
httpd:latest                                             305fd8326a27                      177MB           47.6MB
lancachenet/ubuntu:latest                              f86a1b63836e                      248MB           61.4MB
mysql:8.0                                                7dcddc01f13b                      1.1GB           249MB
nginx-app:v1                                             9d124c9d5add                      238MB           63.1MB

```



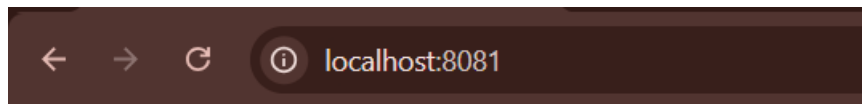
Version 1

Hello from CloudEthiX

-Update the source code file

```
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX</title>
</head>
<body>
  <h1>Version 2</h1>
  <h2>Application Updated Successfully</h2>
</body>
</html>
~
~
~
~
~
~
~
~
~
~
~
~
```

-access the application through the browser



Version 2

Application Updated Successfully

34. **Bridge network:** Create two Docker containers using different images, such as NGINX and MySQL. Connect them using a user-defined bridge network and ensure they can communicate with each other.

-create two docker containers using different images such as NGINX and MySQL

```
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
d2e110a3dd5d   nginx    "/docker-entrypoint..." 3 minutes ago Up 3 minutes    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp    nginx-web
d6f0c28eb95a   mysql:8.0 "docker-entrypoint.s..." 13 minutes ago Up 13 minutes    3306/tcp, 33060/tcp                    mysql-db
root@LAPTOP-HPUOTJF1:~# docker images
IMAGE          ID              DISK USAGE  CONTENT SIZE  EXTRA
mysql:8.0      7dcddc01f13b   1.1GB       249MB         U
nginx:latest   5a88c9c45479   241MB       66MB          U
root@LAPTOP-HPUOTJF1:~#
```

- ensure they can communicate with each other.

```

root@LAPTOP-HPUOTJF1:~# docker network inspect cloud-network
[
  {
    "Name": "cloud-network",
    "Id": "a964394174fa54a200110e17d82484e7b10a72c6b8ffd3f98544bcea482fda69",
    "Created": "2026-07-22T05:59:04.424271266Z",
    "Scope": "local",
    "Driver": "bridge",
    "EnableIPv4": true,
    "EnableIPv6": false,
    "IPAM": {
      "Driver": "default",
      "Options": {},
      "Config": [
        {
          "Subnet": "172.20.0.0/16",
          "IPRange": "",
          "Gateway": "172.20.0.1"
        }
      ]
    },
    "Internal": false,
    "Attachable": false,
    "Ingress": false,
    "ConfigFrom": {
      "Network": ""
    },
    "ConfigOnly": false,
    "Options": {},
    "Labels": {},
    "Containers": {
      "329cf72a75daa3b20eb1f091777af970068243b6522a9c3639f90900a881913e": {
        "Name": "nginx-web",
        "EndpointID": "d060455ec27b1843f9f7ea3c73fde7d8839da65de544377e35aa2a47fa845037",
        "MacAddress": "9a:b3:63:93:ca:56",
        "IPv4Address": "172.20.0.2/16",
        "IPv6Address": ""
      },
      "432dfd91d22ff977abfa99aba83acb7e16aa1b2bf6be8b883d2b08997c95efc5": {
        "Name": "mysql-db",
        "EndpointID": "b854a5019a19c4152815b040fec5d3bf1ac485dff292eadd462738411f2669b5",
        "MacAddress": "9a:b3:63:93:ca:56",
        "IPv4Address": "172.20.0.3/16",
        "IPv6Address": ""
      }
    }
  }
]

root@LAPTOP-HPUOTJF1:~# docker exec nginx-web getent hosts mysql-db
172.20.0.3      mysql-db
root@LAPTOP-HPUOTJF1:~# docker exec mysql-db getent hosts nginx-web
172.20.0.2      nginx-web
root@LAPTOP-HPUOTJF1:~#

```

35. **Port mapping:** Run a containerized web server (e.g., NGINX or Apache) and map its default port (80 or 443) to a custom port on the host machine. Verify that the web server is accessible through the custom port on the host.

```

mysql:8.0 76cdd01f30 1.1GB 249MB 0
root@LAPTOP-HPUOTJF1:~# docker run -d --name nginx-port -p 8080:80 nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
d26f27cc8c41: Pull complete
cacfcdd01f30: Pull complete
3c7ab7949321: Pull complete
b6698f04e005: Pull complete
2bedaf25031a: Pull complete
82454cdbf456: Pull complete
eaid76ccc2c6: Download complete
6c496f5b5050: Download complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
b92d613b0a34d9df8aecd3d398279cecf3d73abeff2754da1f2e5605e1bedb9
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
b92d613b0a34   nginx    "/docker-entrypoint.s..." 6 seconds ago  Up 5 seconds  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  nginx-port
d6f0c28eb95a   mysql:8.0 "docker-entrypoint.s..." 21 minutes ago Up 21 minutes  3306/tcp, 33060/tcp                mysql-db
root@LAPTOP-HPUOTJF1:~#

```

- Verify that the web server is accessible through the custom port on the host.

```

d6f0c28eb95a mysql:8.0 "docker-entrypoint.s..."
root@LAPTOP-HPUOTJF1:~# docker port nginx-port
80/tcp -> 0.0.0.0:8080
80/tcp -> [::]:8080
root@LAPTOP-HPUOTJF1:~# docker inspect nginx-port
[
  {
    "NetworkSettings": {
      "SandboxID": "a19966f277b01ded172f99cb1aea641410945c8fdf1aeba577649965d2c2ff8a",
      "SandboxKey": "/var/run/docker/netns/a19966f277b0",
      "Ports": {
        "80/tcp": [
          {
            "HostIp": "0.0.0.0",
            "HostPort": "8080"
          },
          {
            "HostIp": "::",
            "HostPort": "8080"
          }
        ]
      }
    }
  }
]

```

36. **Host networking:** Run a container with the host networking mode and compare its network configuration with that of the host machine. Explain the advantages and disadvantages of using host networking for containers.

- Run a container with the host networking mode and compare its network configuration

```

root@LAPTOP-HPUOTJF1:~# docker run -d --name nginx-host --network host nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
d26f27cc8c41: Pull complete
2bedaf25031a: Pull complete
b6698f04e005: Pull complete
cacfcdd01f30: Pull complete
3c7ab7949321: Pull complete
82454cdbf456: Pull complete
6c496f5b5050: Download complete
ea1d76ccc2c6: Download complete
Digest: sha256:5a88c9c45479443d7be2eadc894b4ed0a9801bae03d97a5760ae13b5c2005942
Status: Downloaded newer image for nginx:latest
44c64692a9a63fe9c9563fe5b5fa6c62bffb9a396d6f3fd2d51190affe611556
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS    NAMES
44c64692a9a6   nginx    "/docker-entrypoint...." 4 seconds ago    Up 3 seconds    nginx-host
root@LAPTOP-HPUOTJF1:~# hostname -I
172.18.178.20 172.17.0.1 172.19.0.1 172.20.0.1 172.21.0.1 172.22.0.1
root@LAPTOP-HPUOTJF1:~# |

```

- Check the container's network configuration

```

exit
root@LAPTOP-HPUOTJF1:~# docker exec -it nginx-host /bin/bash
root@LAPTOP-HPUOTJF1:/# hostname -I
172.18.178.20 172.17.0.1 172.19.0.1 172.20.0.1 172.21.0.1 172.22.0.1
root@LAPTOP-HPUOTJF1:/# |

```

Advantages of Host Networking

1. Better Performance

- No Network Address Translation (NAT) is used.
 - Packets travel directly through the host's network stack, resulting in lower latency and higher throughput.
2. **No Port Mapping Required**
 - You don't need to use the -p or --publish option.
 - Applications running in the container are directly accessible through the host's network.
 3. **Direct Access to Host Network**
 - The container can use the host's network interfaces and IP address.
 - Useful for applications that need direct access to the network.
 4. **Suitable for High-Performance Applications**
 - Commonly used for network monitoring tools, proxies, load balancers, and applications where network performance is critical.

Disadvantages of Host Networking

1. **No Network Isolation**
 - The container shares the host's network namespace.
 - It does not have its own separate IP address.
2. **Port Conflicts**
 - If the host is already using a port (for example, port 80), another container using host networking cannot use the same port.
 - Only one application can bind to a given port on the host.
3. **Reduced Security**
 - Since the container shares the host's network, it has greater access to the host's networking resources.
 - This increases the potential impact if the container is compromised.
4. **Less Flexible**
 - Host networking is not ideal when running multiple instances of the same application on the same machine because they cannot listen on the same ports.

37. **Named volume:** Launch a containerized database (e.g., MySQL or PostgreSQL) using a named volume to store its data. Stop and remove the container, then recreate it using the same named volume. Verify that the data persists across container recreations.

-Launch a containerized database using named volume to store its data

```

root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-db -v mysql-data:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -e MYSQL_DATABASE=cloudethix -p 3307:3306 mysql:8.0
Unable to find image 'mysql:8.0' locally
8.0: Pulling from library/mysql
4e8a3e0d4e4b: Pull complete
6ef6c7b50a93: Pull complete
a63160a5eda1: Pull complete
0d74d296605b: Pull complete
e3e5d1ac74c1: Pull complete
edf85873f64e: Pull complete
297004cfe470: Pull complete
7534d1db9f8d: Pull complete
49ec2dab01d9: Pull complete
ab24264a27e9: Pull complete
96d30d9fbee8: Pull complete
796812c73292: Download complete
af16638764dd: Download complete
Digest: sha256:7dcddc0f13bab2f15cde676d44d01f61fc9f99fe7785e86196dfc07d358ae2b
Status: Downloaded newer image for mysql:8.0
d302b015ae88de35f5cf384e9e6c62f722d9b6d1a84d8957c021c26835bd4de9
root@LAPTOP-HPUOTJF1:~# docker volume ls
DRIVER      VOLUME NAME
local       5b5bee8723e2e7d40809004e19742c1926190b05cb52bf2ac6dc6b579aa743ce2
local       cab28f1de152a526001218ea642fd77196d83e8780175aff2e52fe311dd2696b
local       cloud-data
local       f4a69a7ecc58d3d714aeac3016399c1ab5d3f01f4e66a031830bf8e8f92fcf0
local       mysql-data

```

local mysql-data

```

root@LAPTOP-HPUOTJF1:~# docker exec -it mysql-db mysql -uroot -p

```

Enter password:

Welcome to the MySQL monitor. Commands end with ; or \g.

Your MySQL connection id is 8

Server version: 8.0.46 MySQL Community Server - GPL

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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```

mysql> USE cloudethix;

```

Database changed

```

mysql> CREATE TABLE employee (

```

```

->   id INT PRIMARY KEY,

```

```

->   name VARCHAR(50)

```

```

-> );

```

Query OK, 0 rows affected (0.03 sec)

```

mysql> INSERT INTO employee VALUES (1,'Achal');

```

Query OK, 1 row affected (0.01 sec)

```

mysql> SELECT * FROM employee;

```

```

+----+-----+
| id | name |
+----+-----+

```

```

| 1 | Achal |
+----+-----+

```

```

1 row in set (0.00 sec)

```

-stop and remove the container

```

Bye
root@LAPTOP-HPUOTJF1:~# docker stop mysql-db
mysql-db
root@LAPTOP-HPUOTJF1:~# docker rm mysql-db
mysql-db
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-db-new -v mysql-data:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -e MYSQL_DATABASE=cloudethix -p 3307:3306 mysql:8.0
8f03e78845095241703bd56e49c7265cc06af14677b9c7615a4c8bcee8614c6d
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
8f03e7884509   mysql:8.0 "/docker-entrypoint.s..." 6 seconds ago Up 6 seconds 33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp  mysql-db-new
467f1cae5567   nginx    "/docker-entrypoint..." About an hour ago Up About an hour 80/tcp                               nginx-host
b92d613b0a34   nginx    "/docker-entrypoint..." 2 hours ago   Up 2 hours    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  nginx-port
root@LAPTOP-HPUOTJF1:~#

```

-recreate it using a same name volume

```

mysql-db
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-db-new -v mysql-data:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -e MYSQL_DATABASE=cloudethix -p 3307:3306 mysql:8.0
8f03e78845095241703bd56e49c7265cc06af14677b9c7615a4c8bcee8614c6d
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
8f03e7884509   mysql:8.0 "/docker-entrypoint.s..." 6 seconds ago Up 6 seconds 33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp  mysql-db-new
467f1cae5567   nginx    "/docker-entrypoint..." About an hour ago Up About an hour 80/tcp                               nginx-host
b92d613b0a34   nginx    "/docker-entrypoint..." 2 hours ago   Up 2 hours    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  nginx-port
root@LAPTOP-HPUOTJF1:~#

```

- Verify that the data persists across container recreations.


```

root@LAPTOP-HPUOTJF1:~# docker exec -it mysql-db-new mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE cloudethix;
Reading table information for completion of table and column names
You can turn off this feature to get a quicker startup with -A

Database changed
mysql> SELECT * FROM employee;
+----+-----+
| id | name |
+----+-----+
|  1 | Achal |
+----+-----+
1 row in set (0.00 sec)

mysql> |

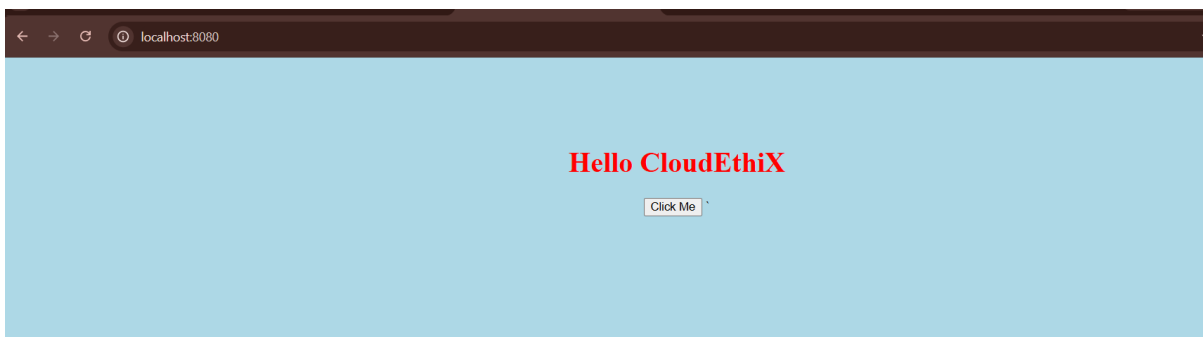
```

38. **Bind mount:** Create a simple web application with a local directory containing HTML, CSS, and JavaScript files. Run a web server container (e.g., NGINX or Apache) and bind mount the local directory to the container's webroot. Verify that the web application is accessible and update the local files to see if changes are reflected in the container.

```

root@LAPTOP-HPUOTJF1:~# docker run -d --name nginx-bind -p 8080:80 -v /root/bind-mount-demo/*:/usr/share/nginx/html nginx
0e35203d4b7c9ae08d082ed22c3b0534e10754ca5d443e91f023a6d6668aee5f
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
0e35203d4b7c   nginx    "/docker-entrypoint. ..."  4 seconds ago    Up 3 seconds    0.0.0.0:8080->80/tcp, [::]:8080->80/tcp    nginx-bind
8f03e7884509   mysql:8.0    "docker-entrypoint.s ..."  22 minutes ago    Up 22 minutes    33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp    mysql-db-new
root@LAPTOP-HPUOTJF1:~# |

```



-update index.html file local file

```

<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Bind Mount Working Successfully</h1>
  <p>Updated without rebuilding the Docker image.</p>

  <button onclick="showMessage()">
    Click Me
  </button>

  <script src="script.js"></script>
</body>
</html>
~
~
~
~
~
~

```

- Verify that the web application is accessible and update the local files to see if changes are reflected in the container.

```
root@LAPTOP-HPUOTJF1:~/bind-mount-demo# vim index.html
root@LAPTOP-HPUOTJF1:~/bind-mount-demo# docker exec -it nginx-bind cat /usr/share/nginx/html/index.html
<!DOCTYPE html>
<html>
<head>
  <title>CloudEthiX</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>
  <h1>Bind Mount Working Successfully</h1>
  <p>Updated without rebuilding the Docker image.</p>

  <button onclick="showMessage()">
    Click Me
  </button>

  <script src="script.js"></script>
</body>
</html>`
root@LAPTOP-HPUOTJF1:~/bind-mount-demo# |
```



39. **Volume backups:** Using a container with a named volume, create a backup of the volume's data by either exporting it as a tarball or copying the data to a host directory. Restore the data to a new container and verify that the restoration was successful.

-Create a container using a name volume

```
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-db -v mysql-data:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -e MYSQL_DATABASE=company -p 3307:3306 mysql:8.0
c5cd6040805937a93734852221dbf17ace8f9cf4b9e51efe0e33ef1d5ac11d88e
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED    STATUS    PORTS                               NAMES
c5cd604080593 mysql:8.0 "docker-entrypoint.s..." 41 seconds ago Up 40 seconds 33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp  mysql-db
012bbee786a4  nginx    "/docker-entrypoint..." 40 minutes ago Up 40 minutes  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  nginx-bind
root@LAPTOP-HPUOTJF1:~# docker exec -it mysql-db mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE company;
EATE TABLE employee(
id INT,
name VARCHAR(50)
);

INSERT INTO employee VALUES(1,'Achal');

SELECT * FROM employee;Database changed
mysql>
mysql> CREATE TABLE employee(
-> id INT,
-> name VARCHAR(50)
-> );
Query OK, 0 rows affected (0.06 sec)

mysql>
mysql> INSERT INTO employee VALUES(1,'Achal');
Query OK, 1 row affected (0.02 sec)

mysql>
mysql> SELECT * FROM employee;
+----+-----+
| id | name |
+----+-----+
|  1 | Achal|
+----+-----+
1 row in set (0.00 sec)

mysql> |
```

-Create a backup dirrectory

```
root@LAPTOP-HPUOTJF1:~# mkdir ~/backup
root@LAPTOP-HPUOTJF1:~# docker run --rm -v mysql-data:/data -v ~/backup:/backup ubuntu tar czvf /backup/mysql_backup.tar.gz -C /data .
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
a3679419df18: Pull complete
ed819469700f: Pull complete
e16351a257e4: Download complete
Digest: sha256:3131b4cc82a783df6c9df078f86e01819a13594b865c2cad47bd1bca2b7063bb
Status: Downloaded newer image for ubuntu:latest
/
```

-copying the data into host dirrectory

```
./performance_schema/replication_appl_167.sdi
./performance_schema/events_statement_127.sdi
./performance_schema/events_stages_hi_112.sdi
./undo_001
./auto.cnf
./ib_buffer_pool
./server-cert.pem
./client-cert.pem
root@LAPTOP-HPUOTJF1:~# ls ~/backup
mysql_backup.tar.gz
root@LAPTOP-HPUOTJF1:~# |
```

- Restore the data to a new container and verify that the restoration was successful.

```
mysql_backup.tar.gz
root@LAPTOP-HPUOTJF1:~# docker stop mysql-db
mysql-db
root@LAPTOP-HPUOTJF1:~# docker rm mysql-db
mysql-db
root@LAPTOP-HPUOTJF1:~# docker volume rm mysql-data
mysql-data
root@LAPTOP-HPUOTJF1:~# docker volume create mysql-data-new
mysql-data-new
root@LAPTOP-HPUOTJF1:~# docker run --rm -v mysql-data-new:/restore -v ~/backup:/backup ubuntu bash -c "cd /restore && tar xzvf /backup/mysql_backup.tar.gz"
./
./undo_002
./binlog.index
./mysql/
./mysql/slow_log.CSM
./mysql/general_log.CSV
./mysql/general_log.CSM
./mysql/slow_log.CSV
./mysql/general_log_213.sdi
```

- Start MySQL using the restored volume

```
./client-cert.pem
root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-new -v mysql-data-new:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -p 3308:3306 mysql:8.0
7ef1b03d5e5538c78af272252821ac5e9af9d012c59b74aee5e45ea89685881
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS        PORTS                               NAMES
7ef1b03d5e55   mysql:8.0 "docker-entrypoint.s..." 4 seconds ago  Up 3 seconds  33060/tcp, 0.0.0.0:3308->3306/tcp, [::]:3308->3306/tcp  mysql-new
012bbe786a4    nginx     "/docker-entrypoint..." About an hour ago  Up About an hour  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  nginx-bind
root@LAPTOP-HPUOTJF1:~# |
```

- Verify the restore

```

root@LAPTOP-HPUOTJF1:~# docker run -d --name mysql-new -v mysql-data-new:/var/lib/mysql -e MYSQL_ROOT_PASSWORD=root123 -p 3308:3306 mysql:8.0
7ef1b03d5e5538c78af272252821ac5e9af9d012c59b74aee5e45eea89685881
root@LAPTOP-HPUOTJF1:~# docker ps
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS                               NAMES
7ef1b03d5e55   mysql:8.0   "docker-entrypoint.s..." 4 seconds ago  Up 3 seconds  33060/tcp, 0.0.0.0:3308->3306/tcp, [::]:3308->3306/tcp  mysql-new
012bbee786a4   nginx       "/docker-entrypoint..." About an hour ago  Up About an hour  0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  nginx-bind
root@LAPTOP-HPUOTJF1:~# docker exec -it mysql-new mysql -uroot -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 8
Server version: 8.0.46 MySQL Community Server - GPL

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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> USE company;
ERROR 1046 (42000): Table 'company' doesn't exist
mysql> CREATE DATABASE company;
mysql> USE company;
mysql> CREATE TABLE employee (
  id INT(4) UNSIGNED ZEROFILL DEFAULT '0000' NOT NULL,
  name VARCHAR(30) DEFAULT '' NOT NULL,
  PRIMARY KEY (id)) ENGINE=InnoDB;
mysql> INSERT INTO employee (id, name) VALUES (1, 'Achal');
mysql> SELECT * FROM employee;
+----+-----+
| id | name |
+----+-----+
|  1 | Achal |
+----+-----+
1 row in set (0.00 sec)

mysql>

```

40. Docker Compose networking: Create a Docker Compose file that defines a multi-container application with a frontend, backend, and database. Set up custom networks and volumes for the services and ensure that they can communicate with each other and store data persistently

-Create a Docker Compose file that defines a multi-container application with a frontend, backend, and database.

```

root@LAPTOP-HPUOTJF1:~#
root@LAPTOP-HPUOTJF1:~# mkdir -p compose-demo/frontend compose-demo/backend
root@LAPTOP-HPUOTJF1:~# ls
Amend_repo      Gitignore-Repo  backup          cloudethix-nginx  githubaction-project  nginx-container.tar  revert_demo      ubuntu-custom
Cloud_Ethix_Copyfiles  Merge_demo      bind-mount-demo  compose-demo      httpd                nginx-update         stash-demo       ubuntu-nginx
DevOps_Project  amend_repo      ci-cd-project   docker-compose-demo  index.html            project              stash1-demo     volume-backup
root@LAPTOP-HPUOTJF1:~# cd compose-demo/
root@LAPTOP-HPUOTJF1:~/compose-demo# ls
backend frontend
root@LAPTOP-HPUOTJF1:~/compose-demo# cd frontend/
root@LAPTOP-HPUOTJF1:~/compose-demo/frontend# vim index.html
root@LAPTOP-HPUOTJF1:~/compose-demo/frontend# cd
root@LAPTOP-HPUOTJF1:~# cd compose-demo/backend/
root@LAPTOP-HPUOTJF1:~/compose-demo/backend# vim package.json
root@LAPTOP-HPUOTJF1:~/compose-demo/backend# vim server.js
root@LAPTOP-HPUOTJF1:~/compose-demo/backend# vim Dockerfile
root@LAPTOP-HPUOTJF1:~/compose-demo/backend# cd ..
root@LAPTOP-HPUOTJF1:~/compose-demo# vim docker-compose.yml
root@LAPTOP-HPUOTJF1:~/compose-demo# docker compose up
WARN[0000] /root/compose-demo/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
WARN[0000] Docker Compose is configured to build using Bake, but buildx isn't installed
[+] Building 62.9s (12/12) FINISHED
=> [backend internal] load build definition from Dockerfile
=> transferring dockerfile: 151B
=> [backend internal] load metadata for docker.io/library/node:20
=> [backend auth] library/node:pull token for registry-1.docker.io
=> [backend internal] load .dockerignore
=> transferring context: 2B

```

-docker-compose file

```

version: "3.9"

services:
  frontend:
    image: nginx
    container_name: frontend
    ports:
      - "8080:80"
    volumes:
      - ./frontend:/usr/share/nginx/html
    networks:
      - cloud-network

  backend:
    build: ./backend
    container_name: backend
    ports:
      - "3000:3000"
    networks:
      - cloud-network

  database:
    image: mysql:8.0
    container_name: mysql-db
    environment:
      MYSQL_ROOT_PASSWORD: root123
      MYSQL_DATABASE: company
    volumes:
      - mysql-data:/var/lib/mysql
    ports:
      - "3307:3306"
    networks:
      - cloud-network

networks:
  cloud-network:

volumes:
  mysql-data:

```

```

backend exited with code 137
root@LAPTOP-HPUOTJF1:~/compose-demo# docker compose up -d
WARN[0000] /root/compose-demo/docker-compose.yml: the attribute 'version' is obsolete, it will be ignored, please remove it to avoid potential confusion
[+] Running 3/3
✔ Container mysql-db   Started
✔ Container frontend   Started
✔ Container backend    Started
root@LAPTOP-HPUOTJF1:~/compose-demo# docker network ls
NETWORK ID          NAME                DRIVER              SCOPE
14516b0dea3         bridge             bridge              local
a964394174fa        cloud-network       bridge              local
dad2a1b8db20        compose-demo_cloud-network bridge              local
947f59718bf2        compose-project_default bridge              local
3411a96dcd47        docker-compose-demo_cloud-network bridge              local
20f475dfbeef        host               host                local
f1eff302c182        none               null                local
root@LAPTOP-HPUOTJF1:~/compose-demo# docker ps
CONTAINER ID        IMAGE               COMMAND              CREATED            STATUS             PORTS                               NAMES
8e57474524d1       nginx              "/docker-entrypoint..." About a minute ago Up 52 seconds      0.0.0.0:8080->80/tcp, [::]:8080->80/tcp frontend
6b1938d9fce3       mysql:8.0          "docker-entrypoint.s..." About a minute ago Up 52 seconds      33060/tcp, 0.0.0.0:3307->3306/tcp, [::]:3307->3306/tcp mysql-db
a5f2d194c5f4       compose-demo-backend "docker-entrypoint.s..." About a minute ago Up 52 seconds      0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp backend

```

- ensure that they can communicate with each other and store data persistently

```

    com.docker.compose.network: "cloud-network",
    com.docker.compose.project: "compose-demo",
    com.docker.compose.version: "2.40.3"
  },
  "Containers": {
    "6b1938d9fce3e8c2c4d0fdc63dd960741edbe9861aff6e2d30f765c8865011ed": {
      "Name": "mysql-db",
      "EndpointID": "ca915b86c7ce4816a110cb0555aa69253a1b3c6322b2b85ac5bdd92fd2d6b632",
      "MacAddress": "3e:a9:8a:3f:45:87",
      "IPv4Address": "172.22.0.4/16",
      "IPv6Address": ""
    },
    "8e57474524d12f0351b316ce48802d42ff376c8f2d106850bf65db26e00023df": {
      "Name": "frontend",
      "EndpointID": "14d941d8b2761451f201a8cd40ffe86e6b01d3162bff0d6dade43b9e90815698",
      "MacAddress": "42:41:60:83:63:11",
      "IPv4Address": "172.22.0.2/16",
      "IPv6Address": ""
    },
    "a5f2d194c5f4774c230dff6234d46315b4f3e1e0b58ee38c3281696ceb56a58d": {
      "Name": "backend",
      "EndpointID": "84903c207e57f5b45d063090887666958f948a1b14867d499872669ec88c4e49",
      "MacAddress": "da:e8:db:20:4f:62",
      "IPv4Address": "172.22.0.3/16",
      "IPv6Address": ""
    }
  },
  "Status": {

```

